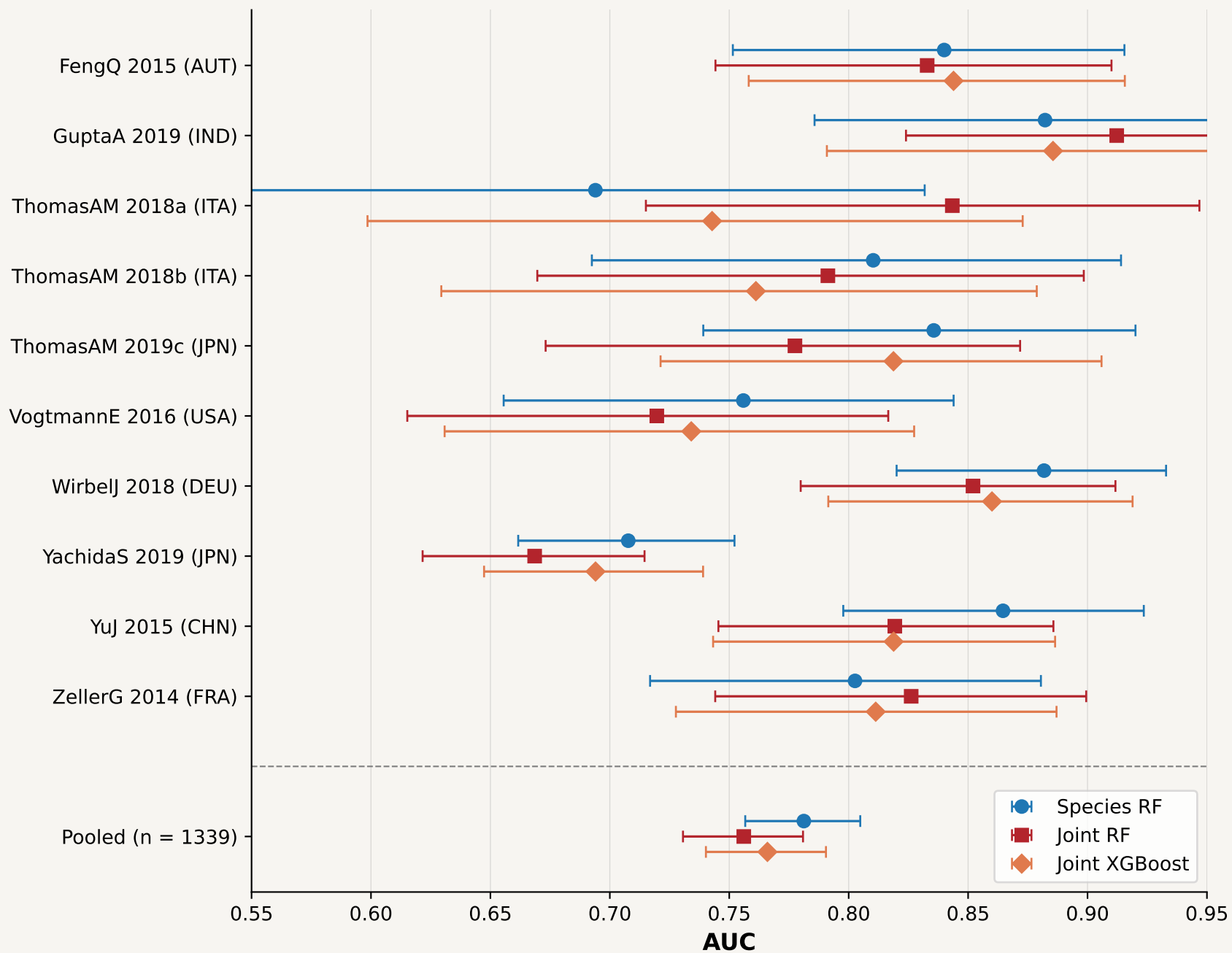
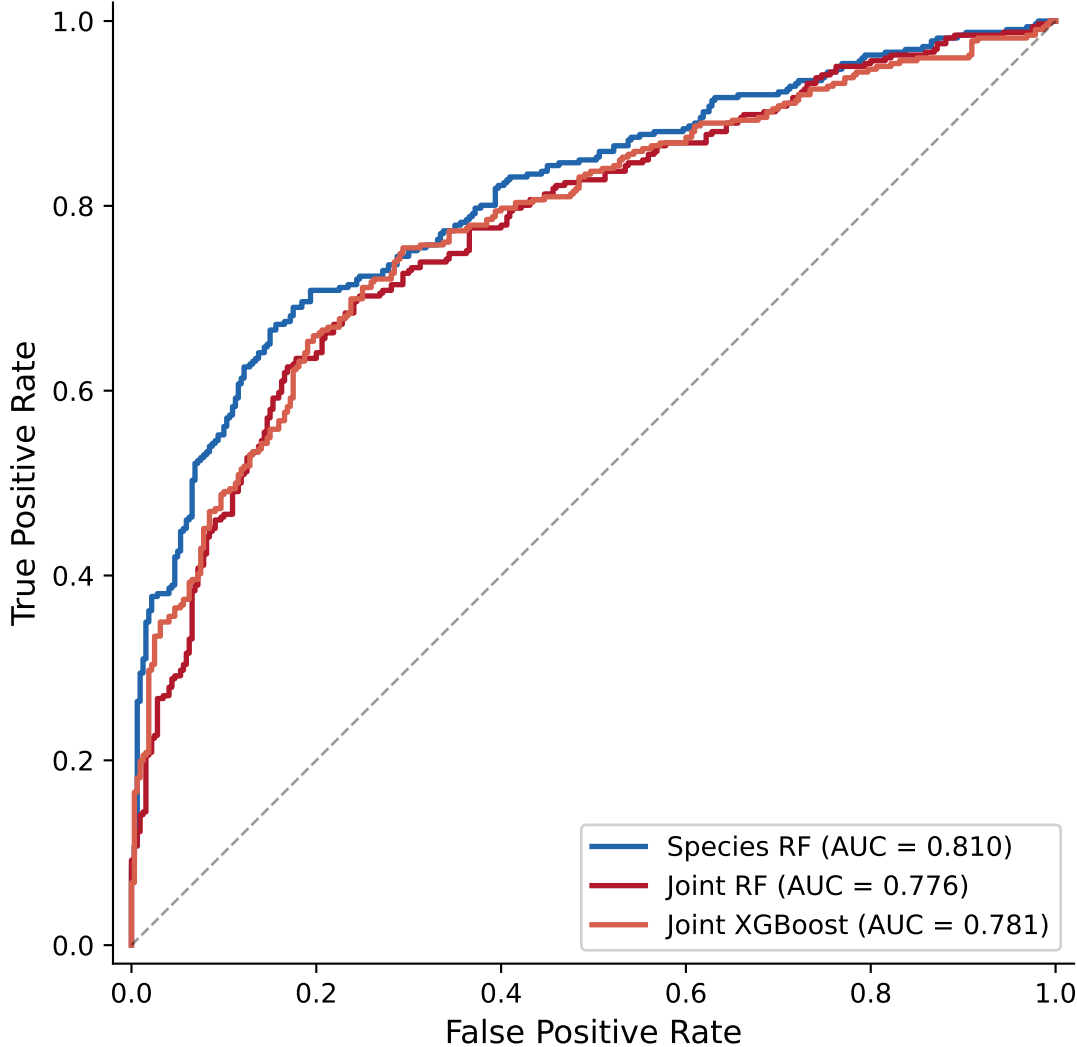


Figure 1. Per-cohort and pooled AUC with 95% bootstrap CI under LODO cross-validation

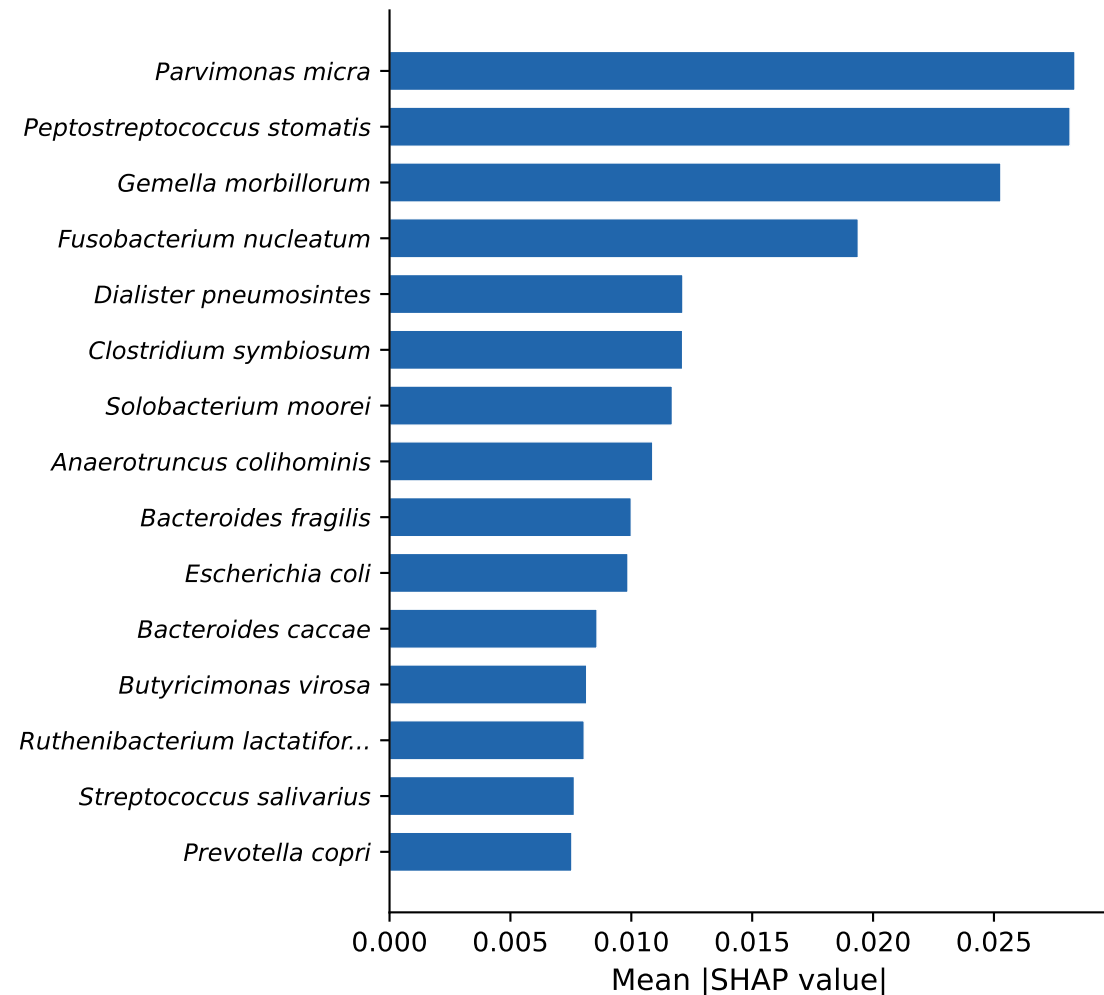


Pooled LODO ROC Curves (n = 646)

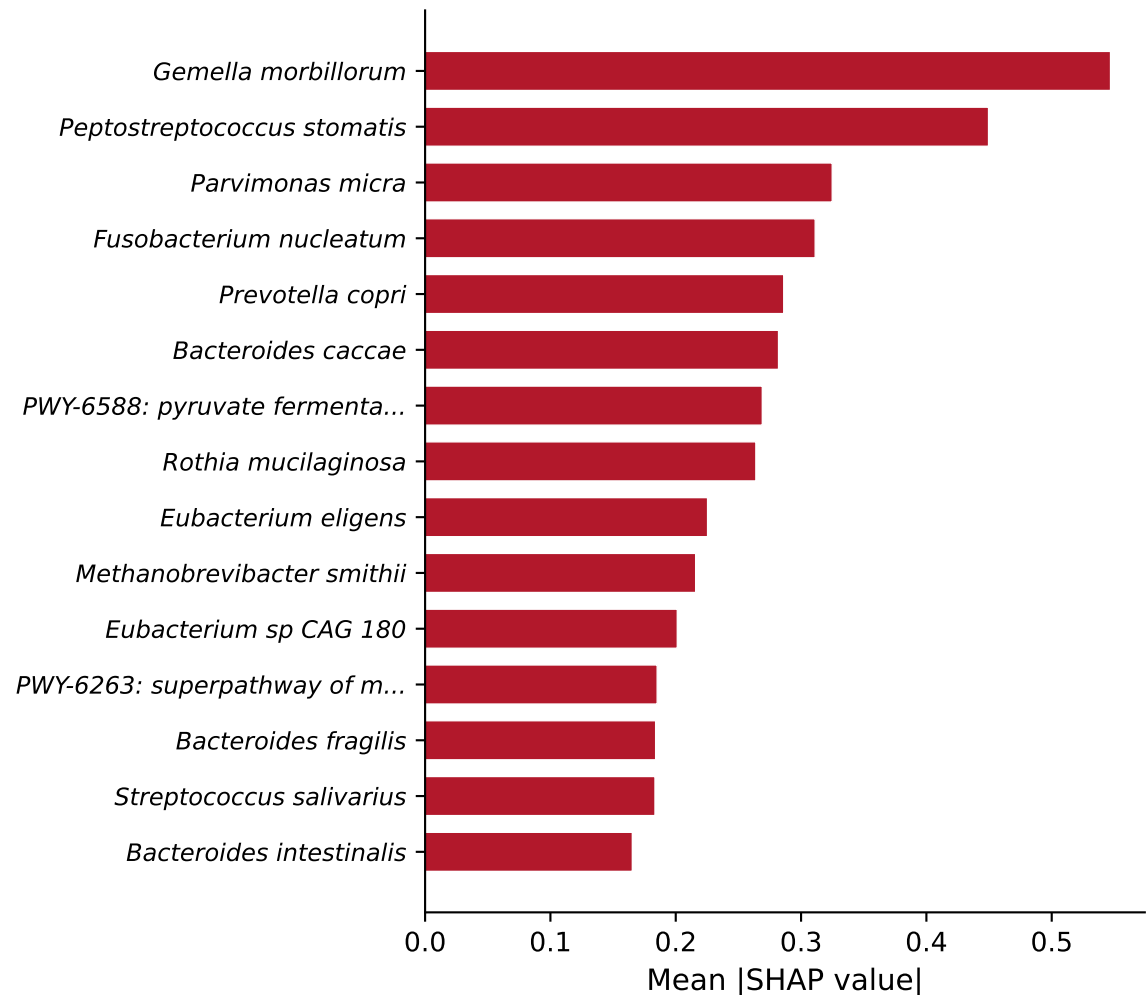


Top 15 Species by SHAP Feature Importance (CRC vs. Healthy)

Random Forest

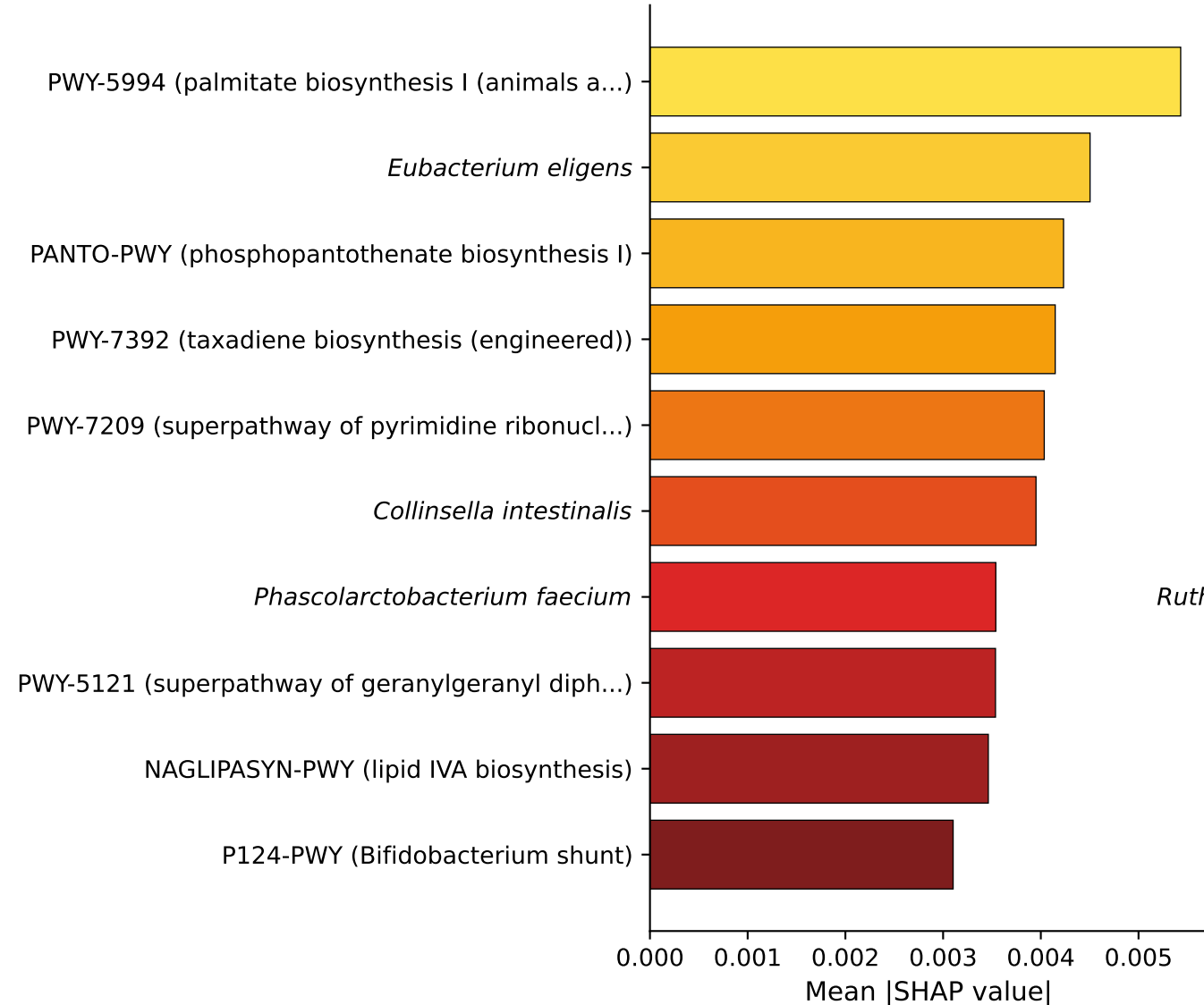


XGBoost

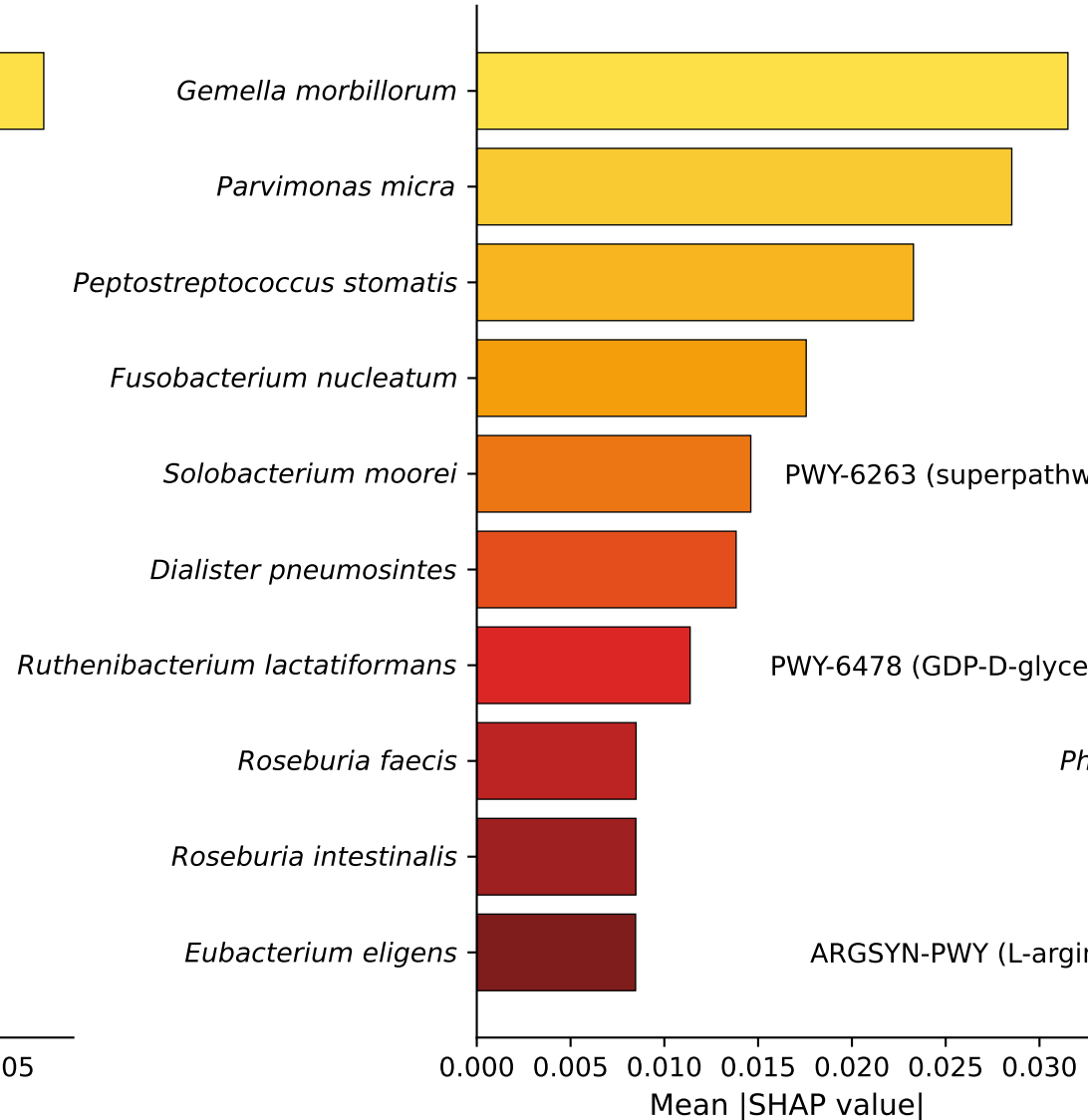


8/15 features shared in top 15 between RF and XGBoost

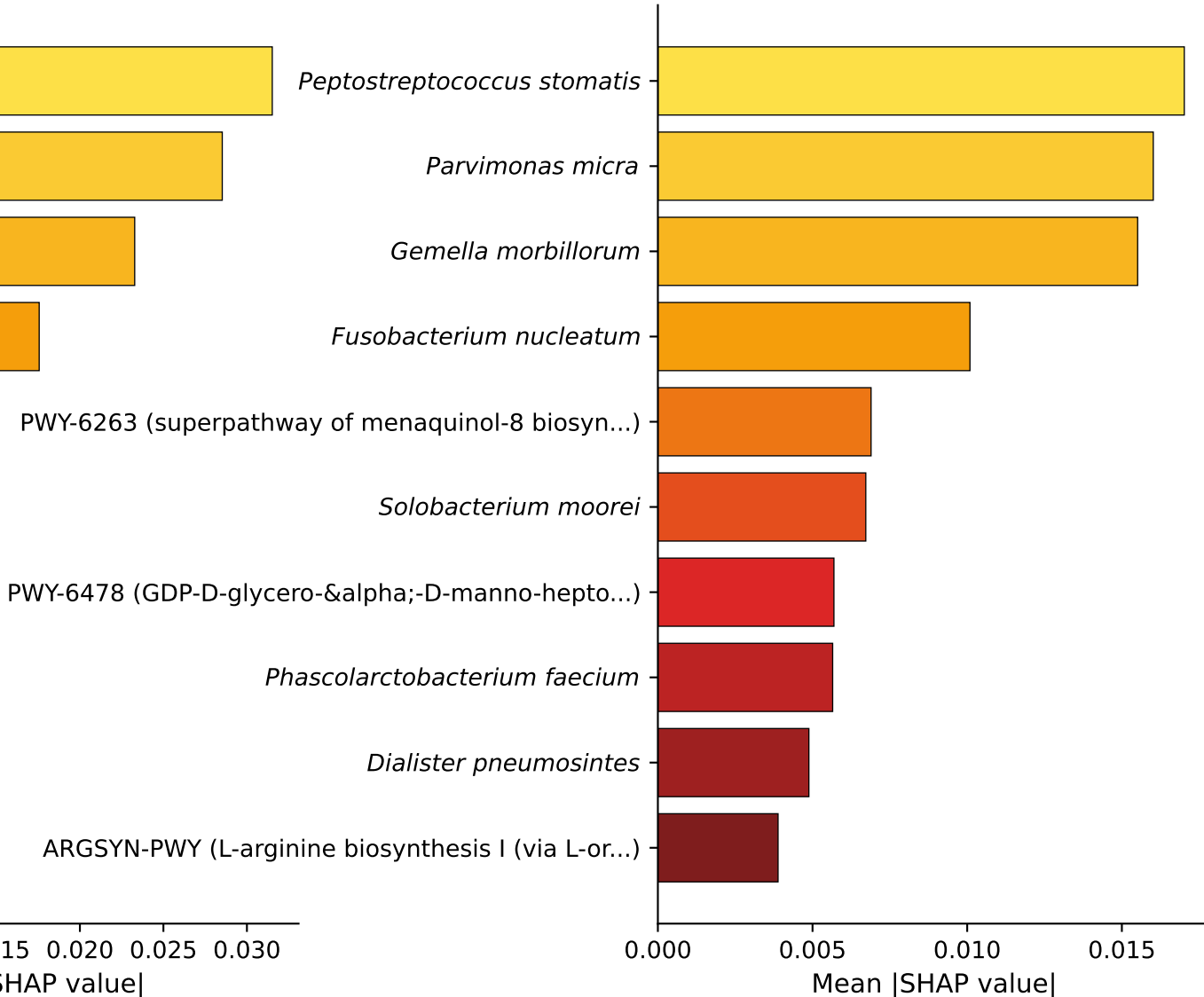
A Healthy vs Adenoma
(early commensal-depletion phase)



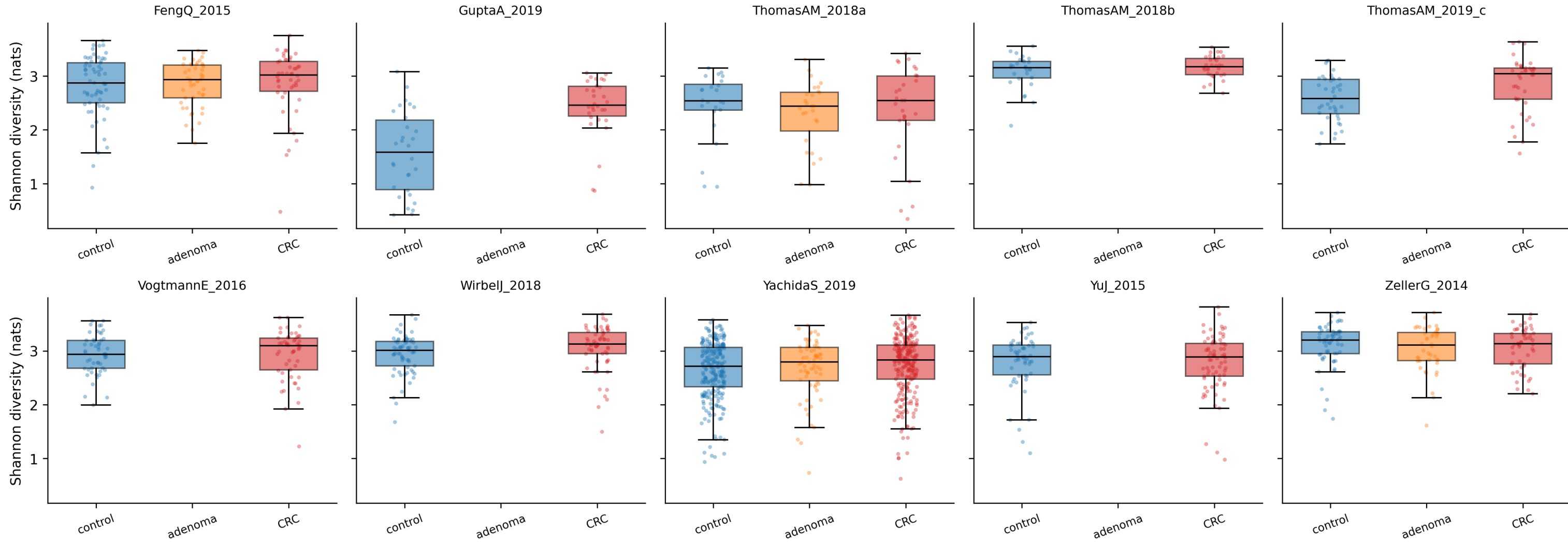
B CRC vs Healthy
(oral pathobiont colonization (full progression))



C Adenoma vs CRC
(oral pathobiont colonization (malignant transition))

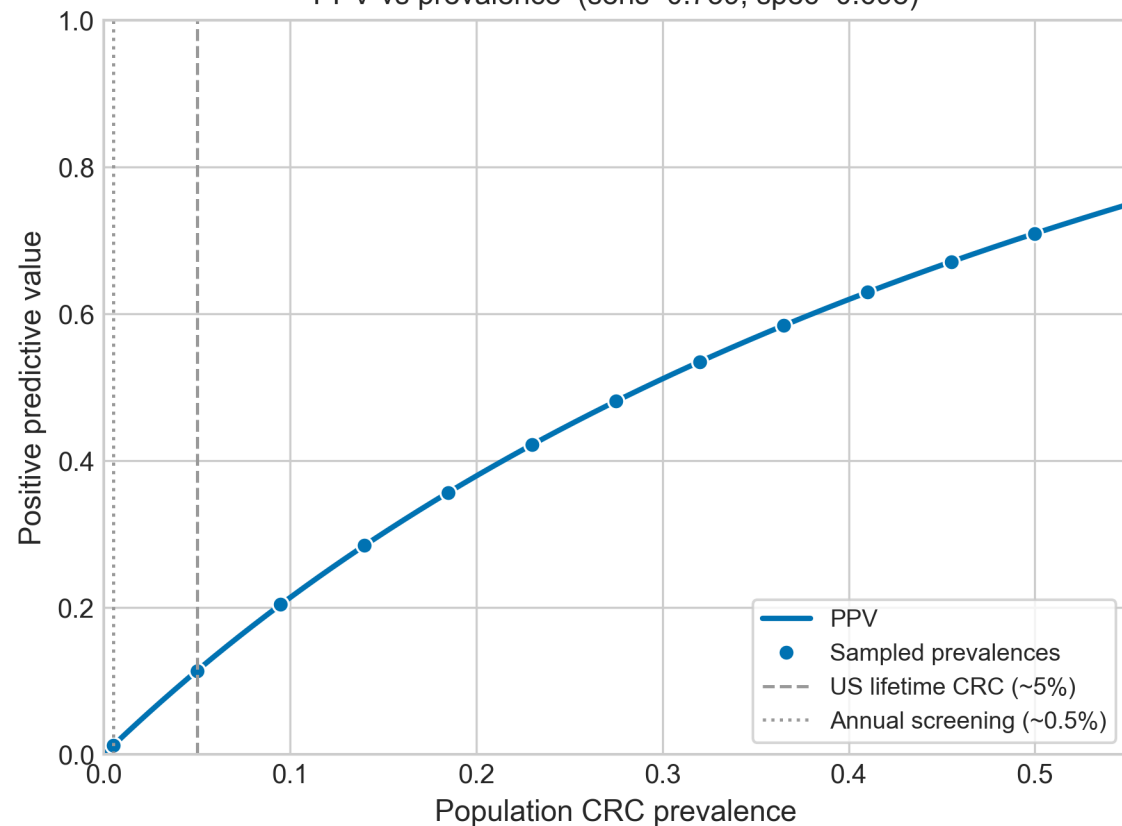


Alpha diversity (Shannon) by study_condition, per cohort

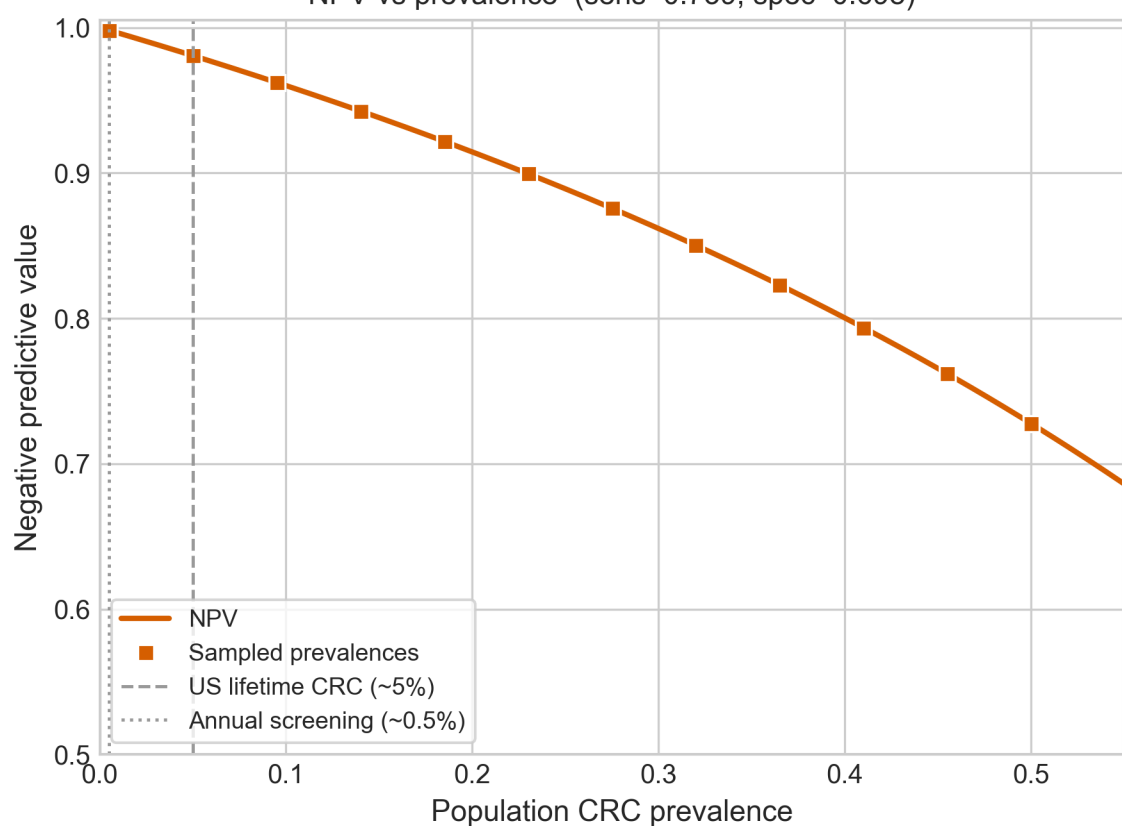


Species RF (Youden-J fixed): base-rate-adjusted PPV and NPV

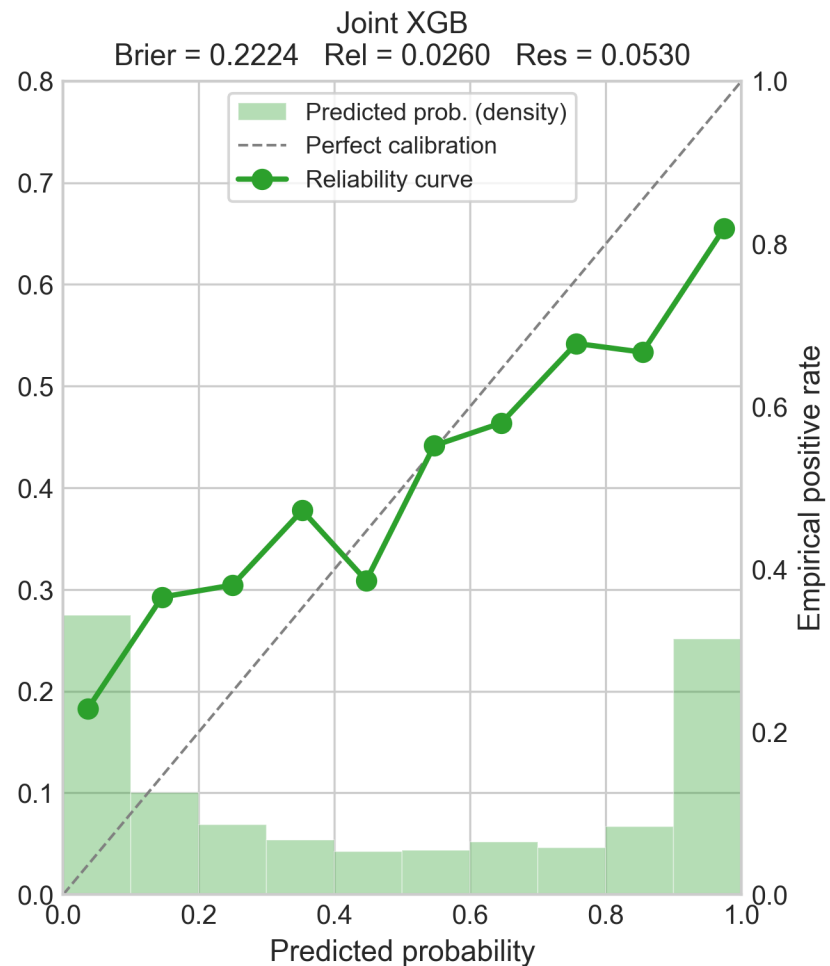
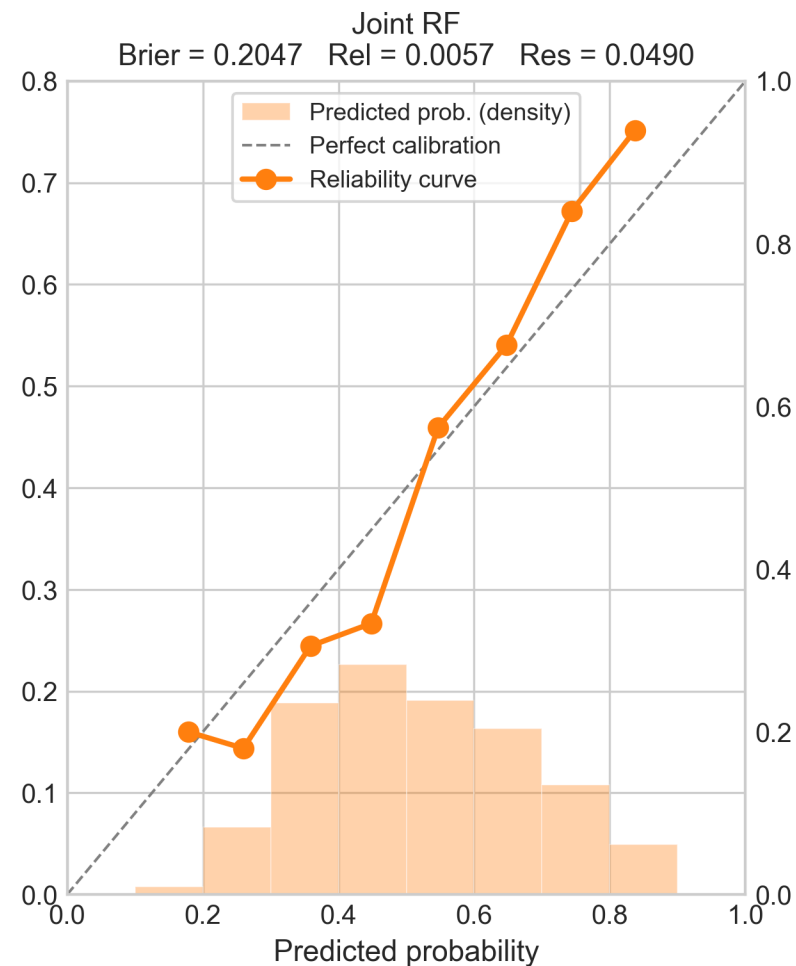
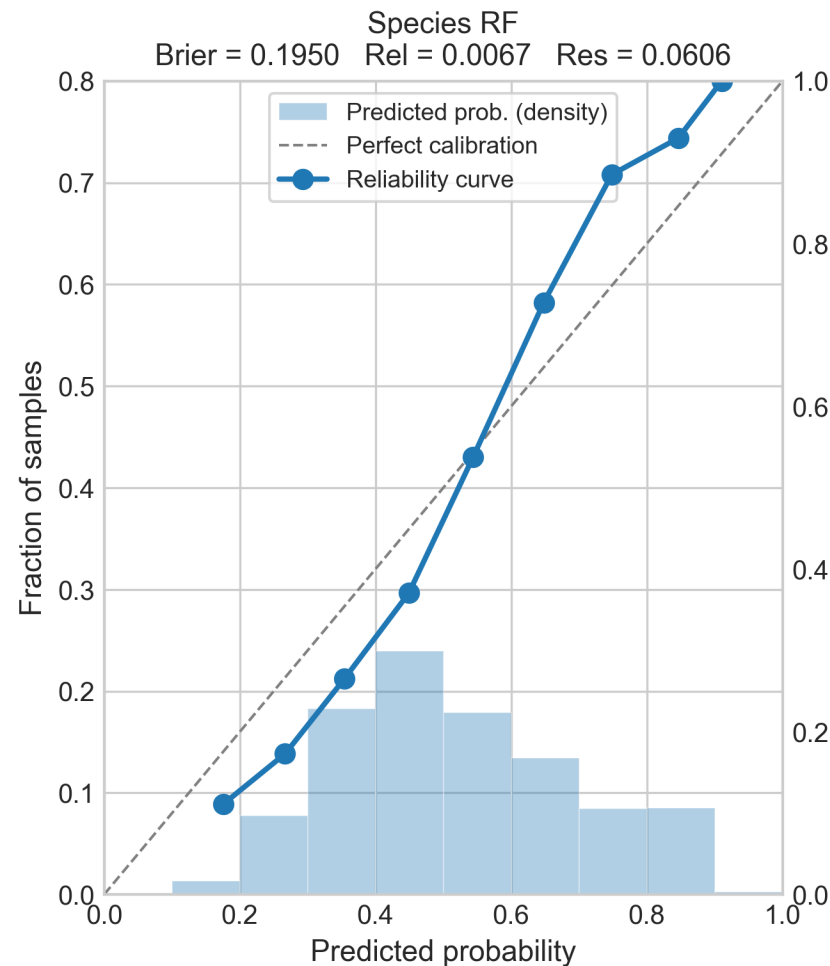
PPV vs prevalence (sens=0.739, spec=0.698)



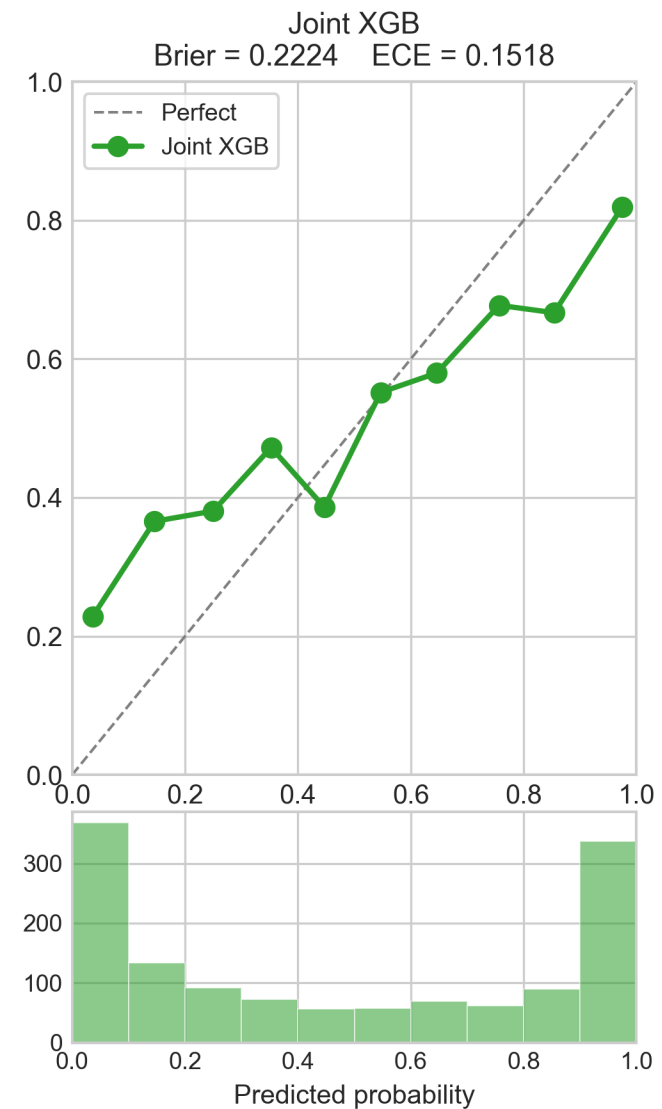
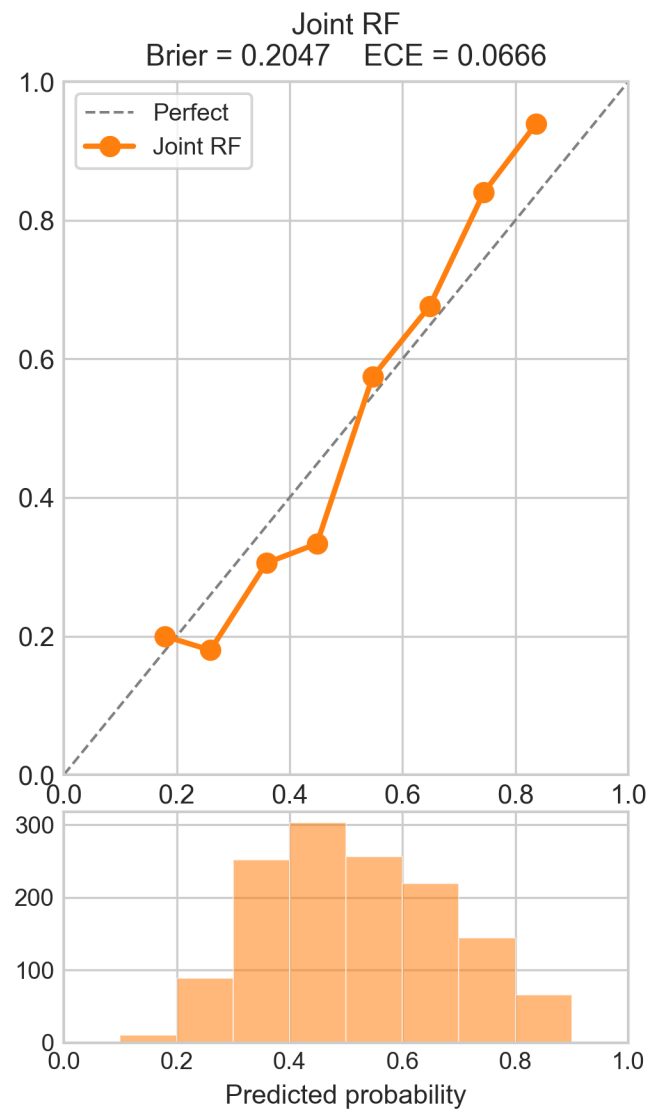
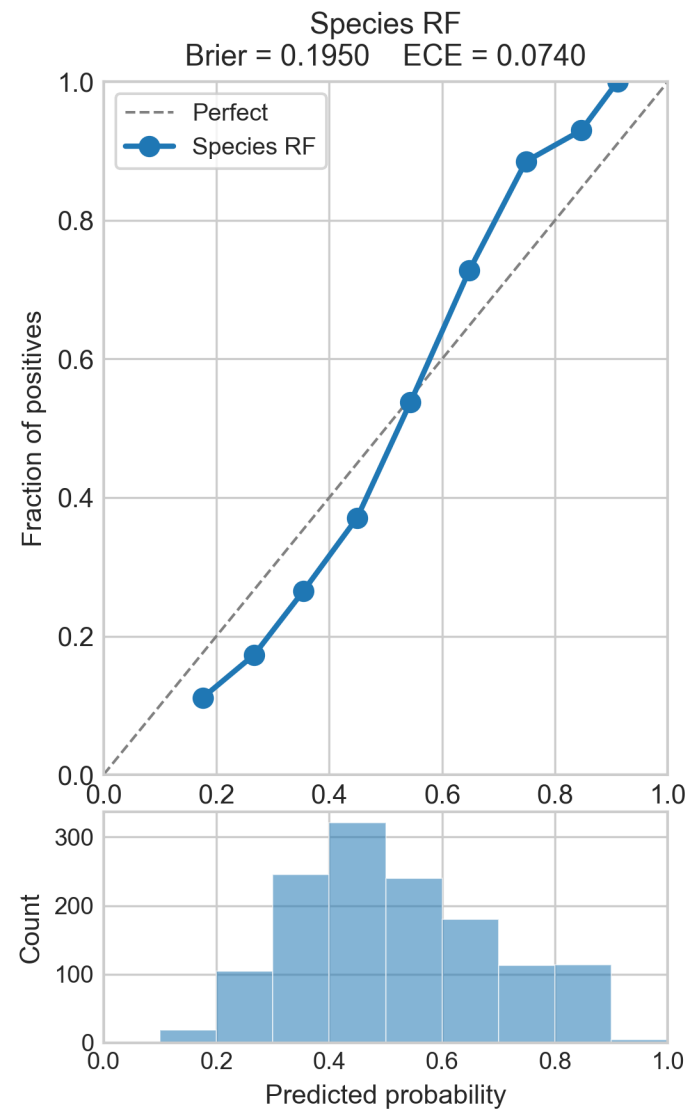
NPV vs prevalence (sens=0.739, spec=0.698)



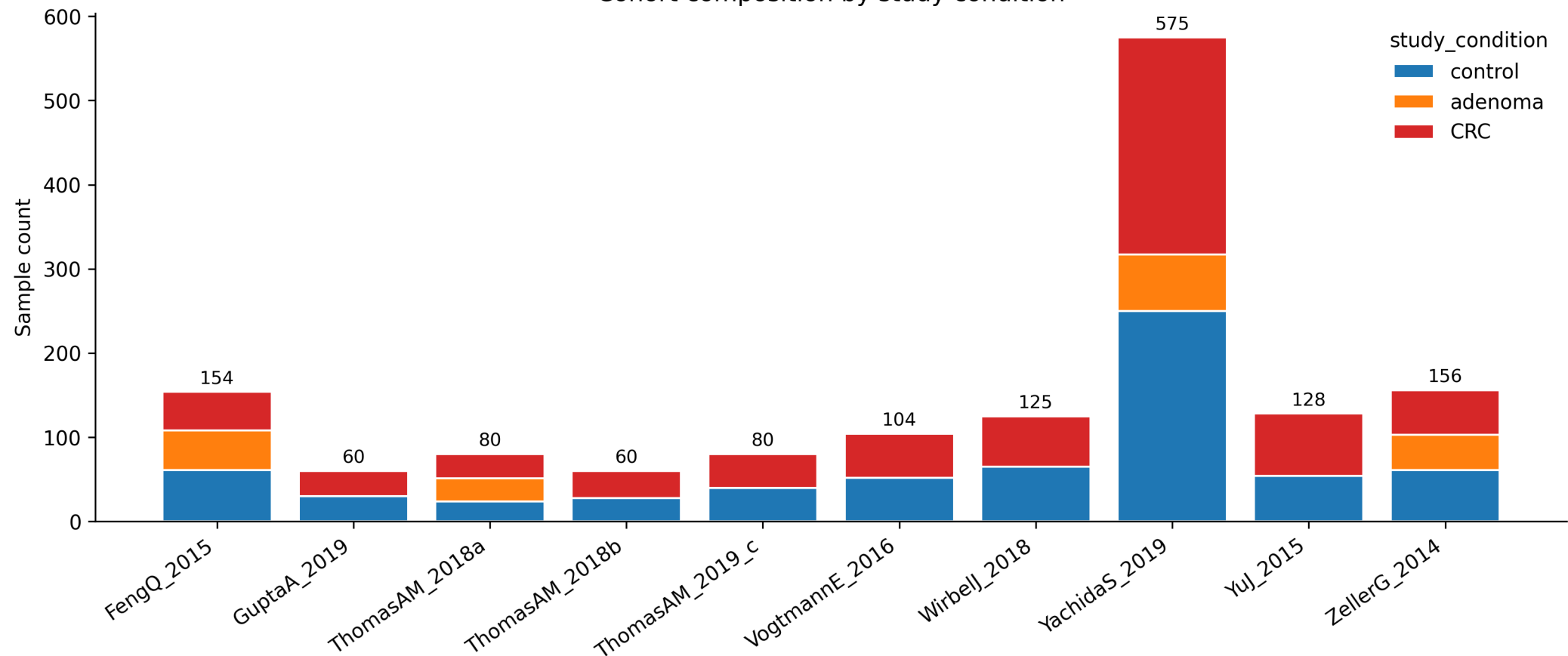
Calibration mechanism — joint XGB pushes probabilities to 0/1, inflating reliability term



Reliability diagrams (10-bin equal-width) — pooled LODO predictions

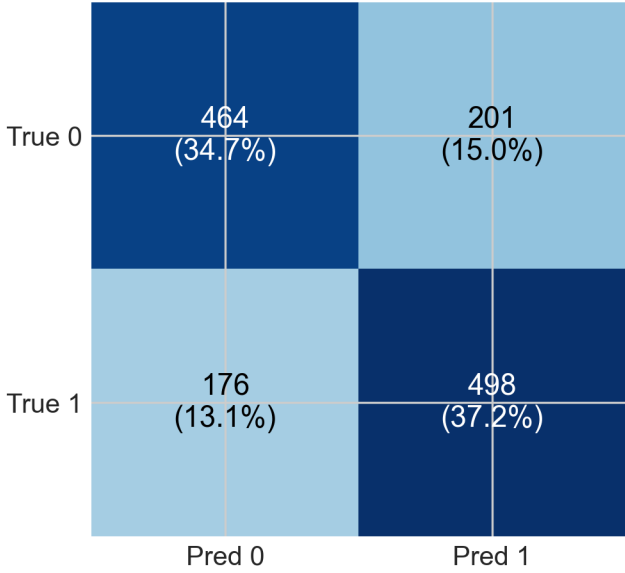


Cohort composition by study condition

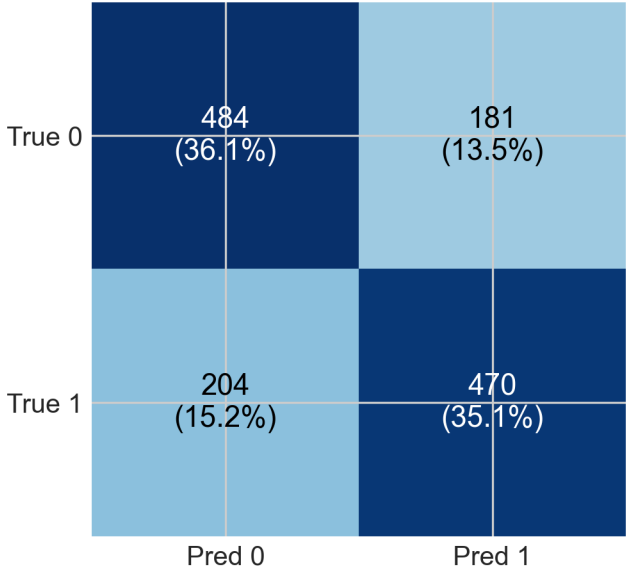


Confusion matrices — pooled LODO predictions

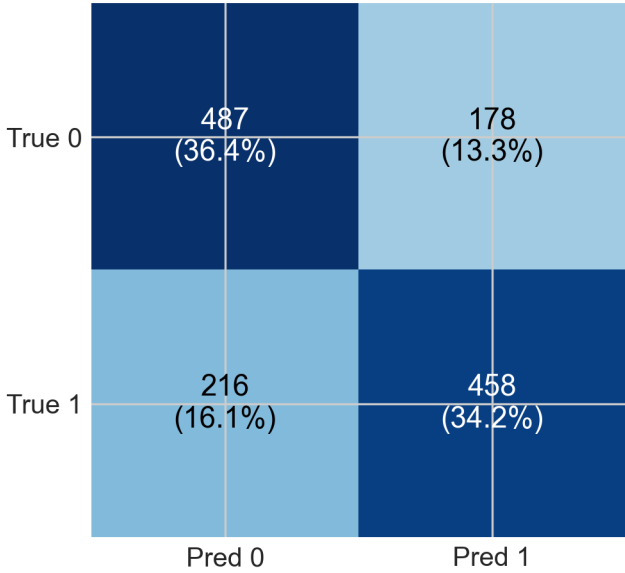
Species RF — Youden-J (t=0.485)
Sens=0.739 Spec=0.698 MCC=0.437



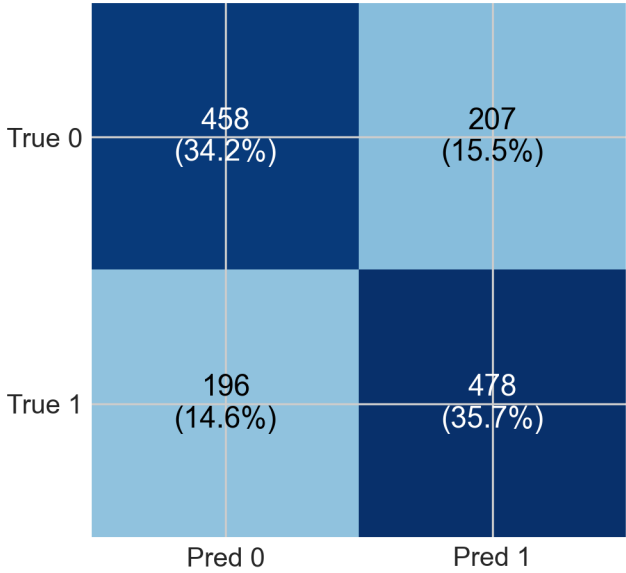
Species RF — Threshold = 0.5 (t=0.500)
Sens=0.697 Spec=0.728 MCC=0.425



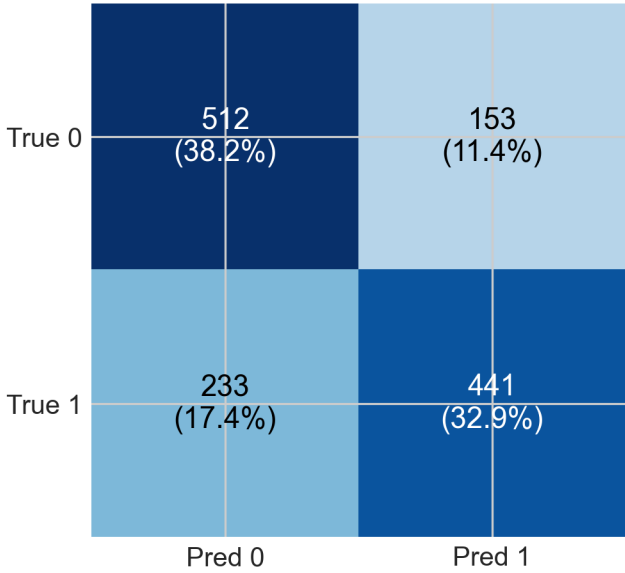
Joint RF — Youden-J (t=0.519)
Sens=0.680 Spec=0.732 MCC=0.412



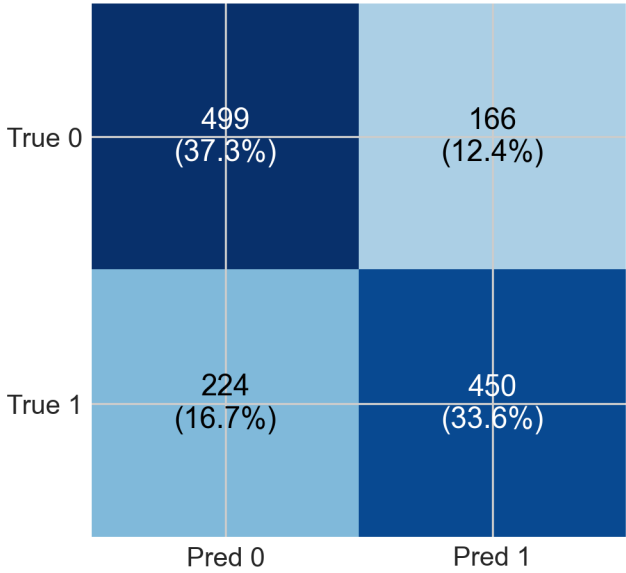
Joint RF — Threshold = 0.5 (t=0.500)
Sens=0.709 Spec=0.689 MCC=0.398



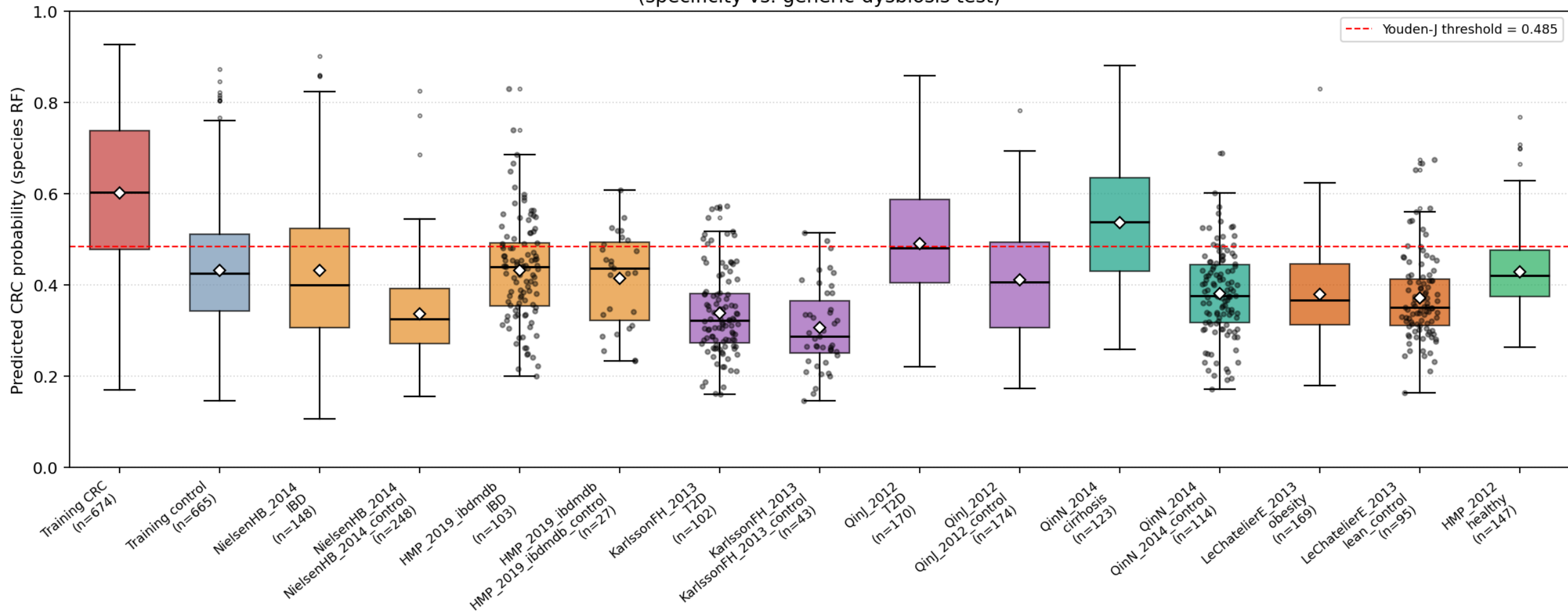
Joint XGB — Youden-J (t=0.533)
Sens=0.654 Spec=0.770 MCC=0.427



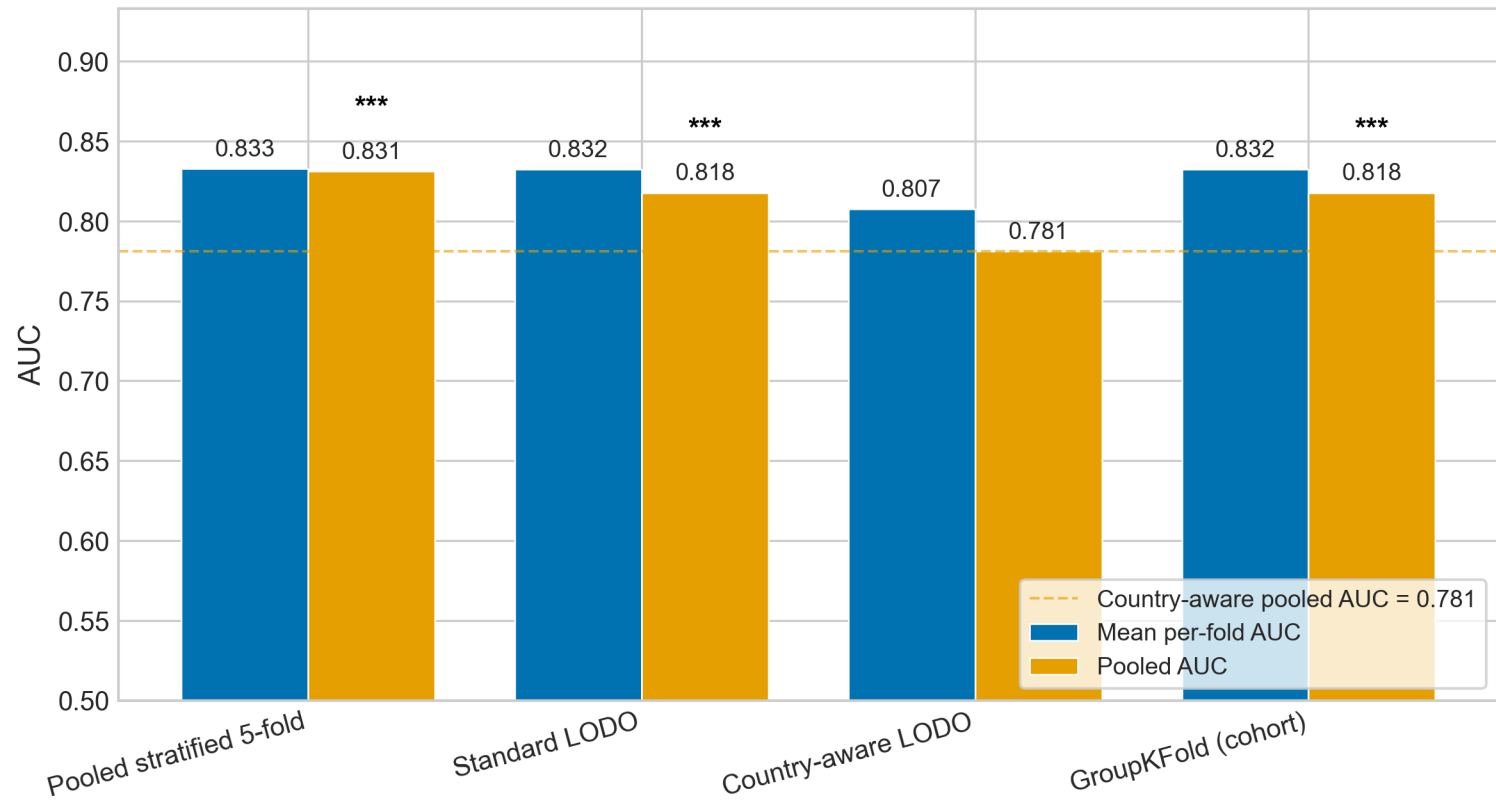
Joint XGB — Threshold = 0.5 (t=0.500)
Sens=0.668 Spec=0.750 MCC=0.419



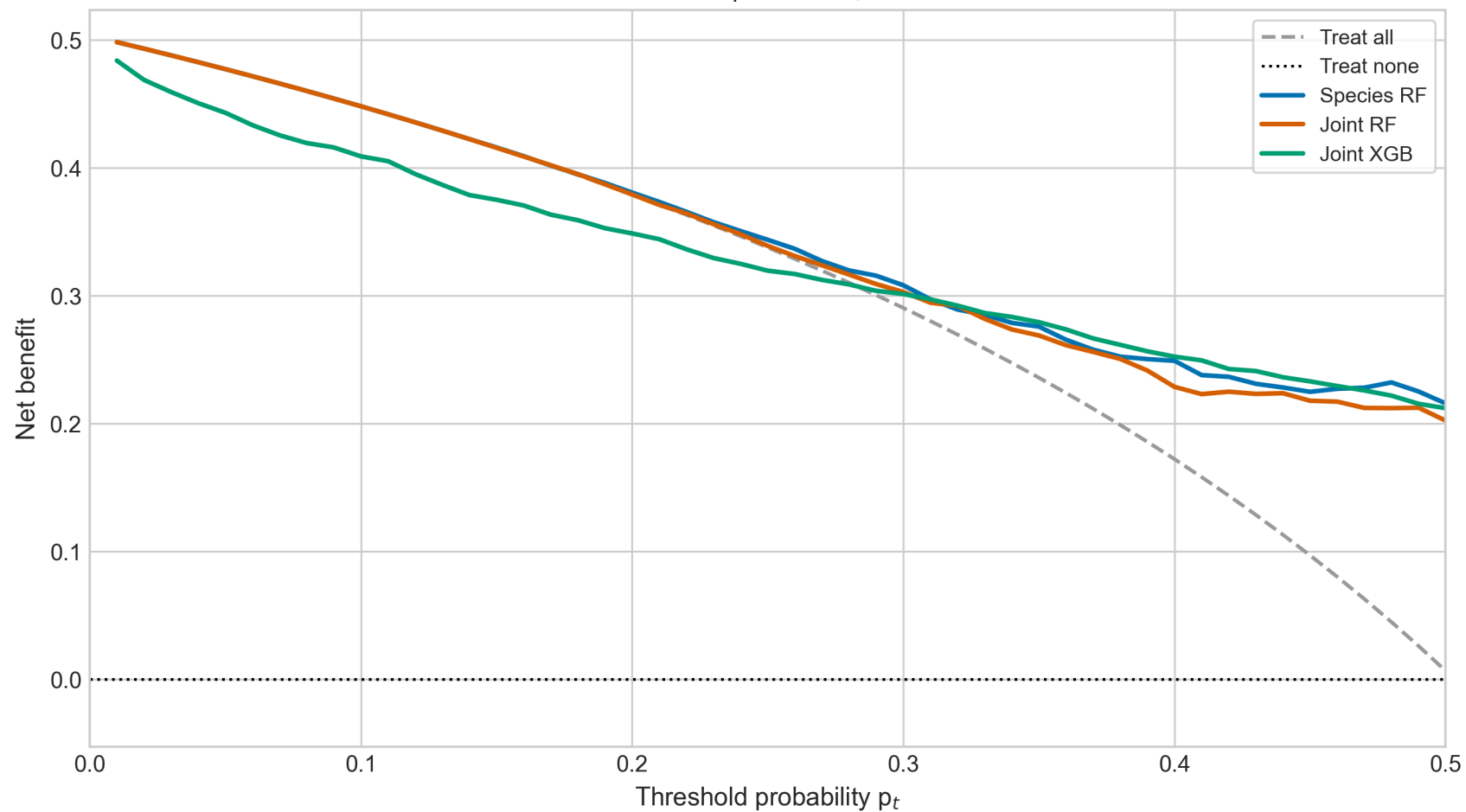
CRC-classifier scores on non-CRC disease cohorts
(specificity vs. generic-dysbiosis test)



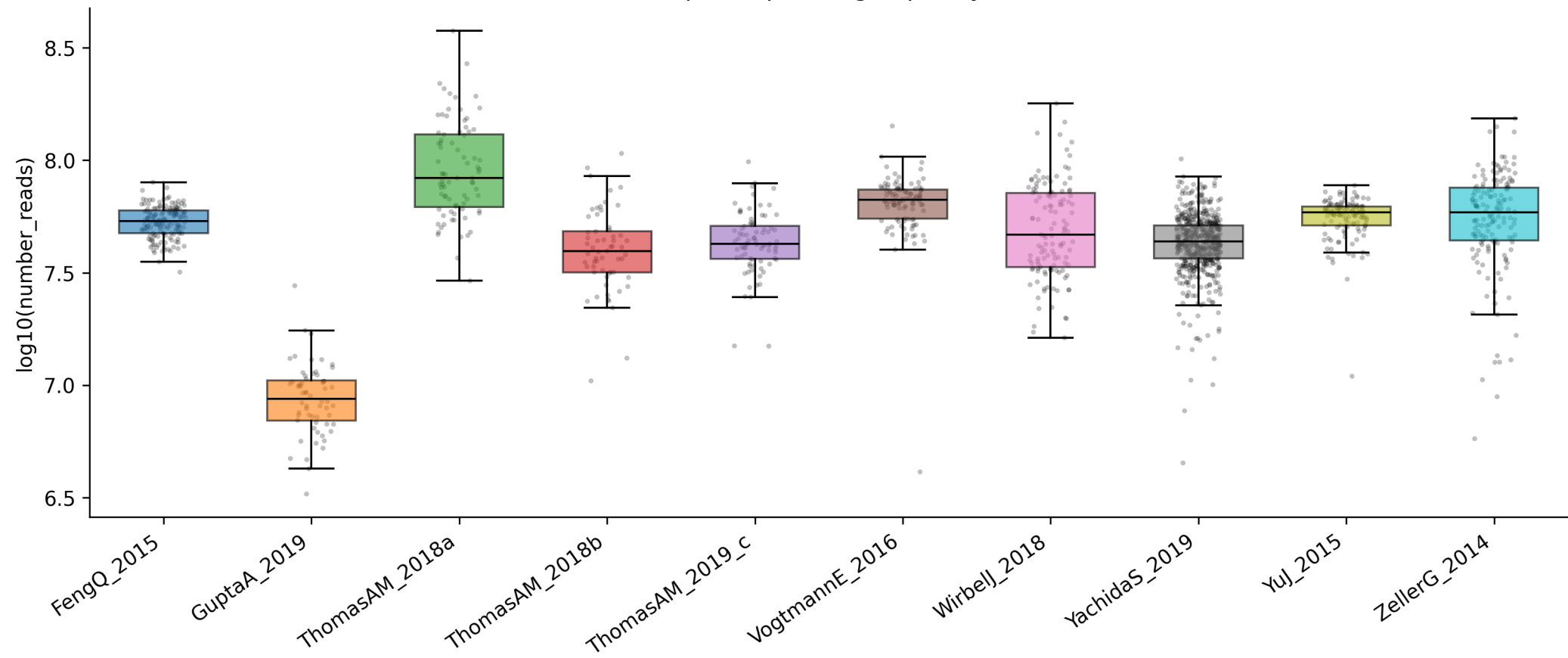
Cross-validation methodology comparison — species RF
(stars: DeLong p vs country-aware LODO on pooled predictions)



Decision-curve analysis (Vickers & Elkin)
Pooled LODO predictions, n = 1339

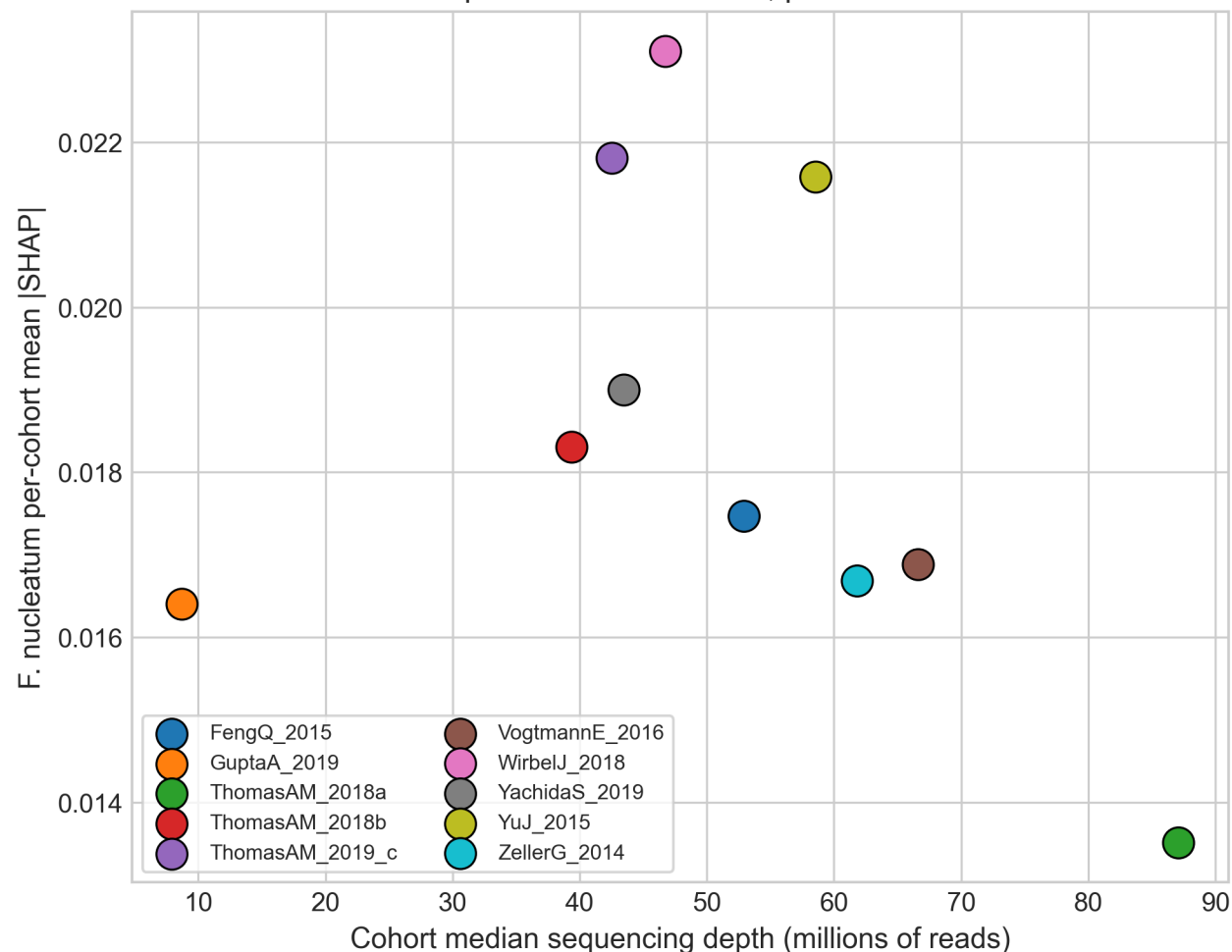


Per-sample sequencing depth by cohort

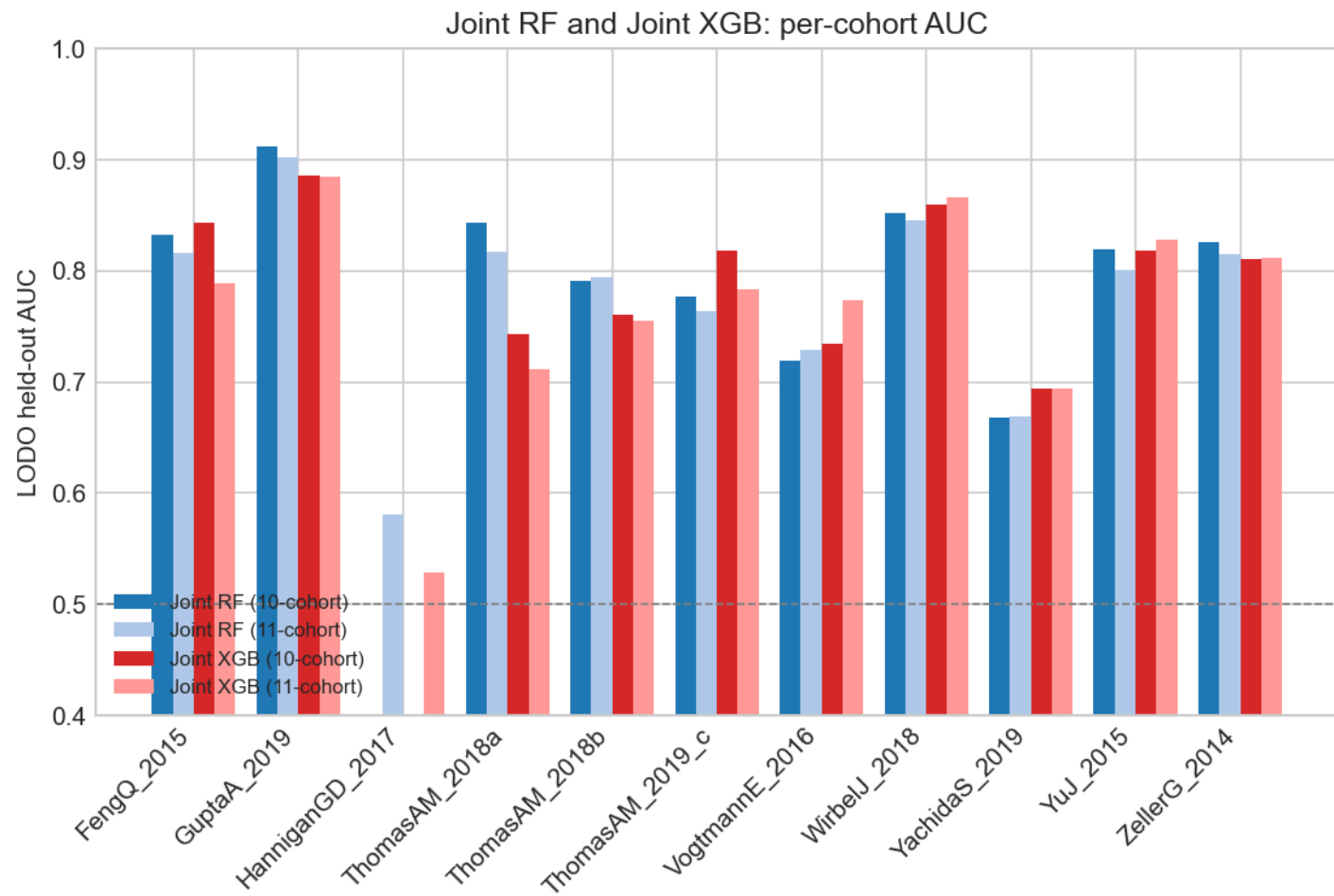
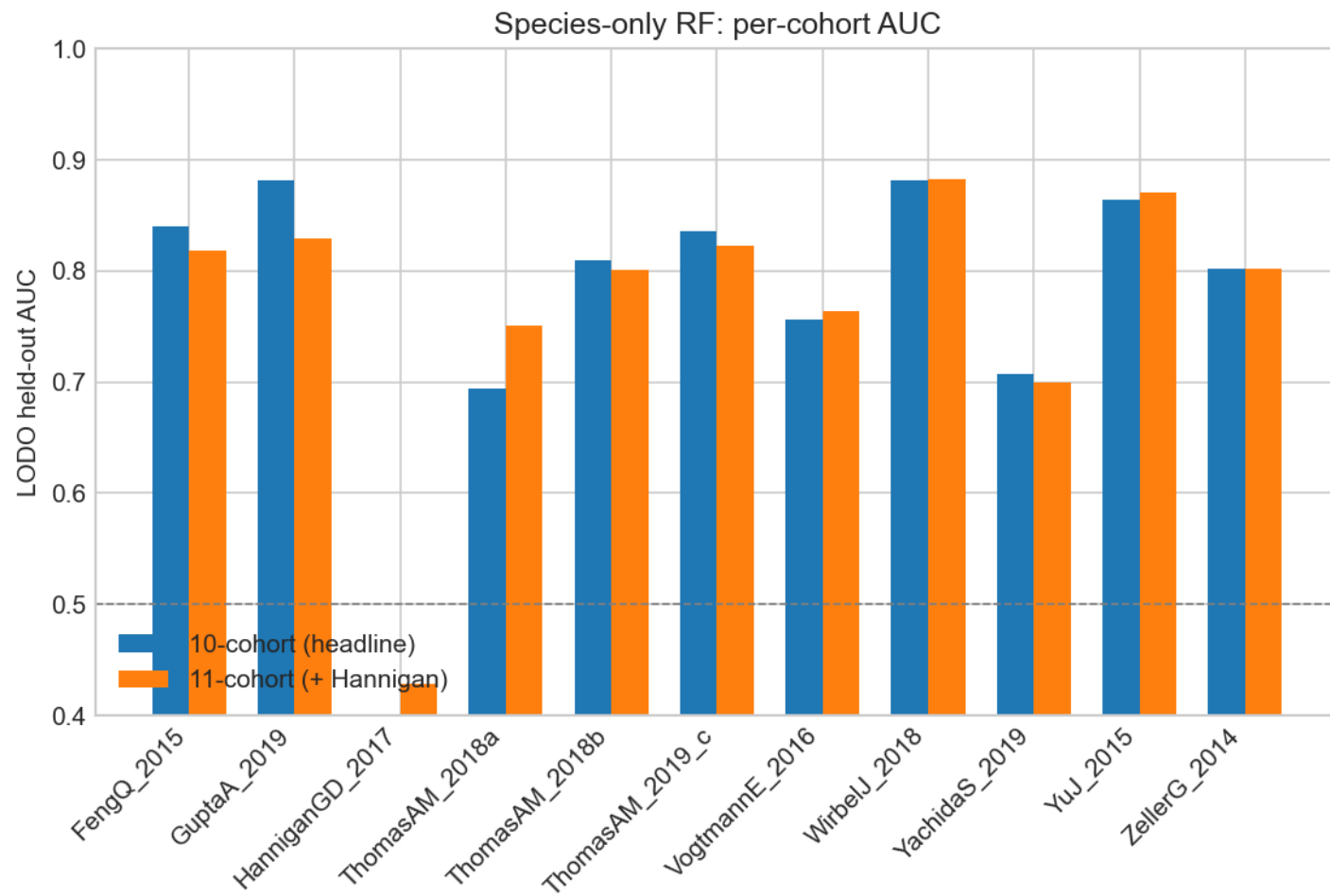


F. nucleatum SHAP vs cohort sequencing depth

Spearman rho = -0.194, p = 0.591

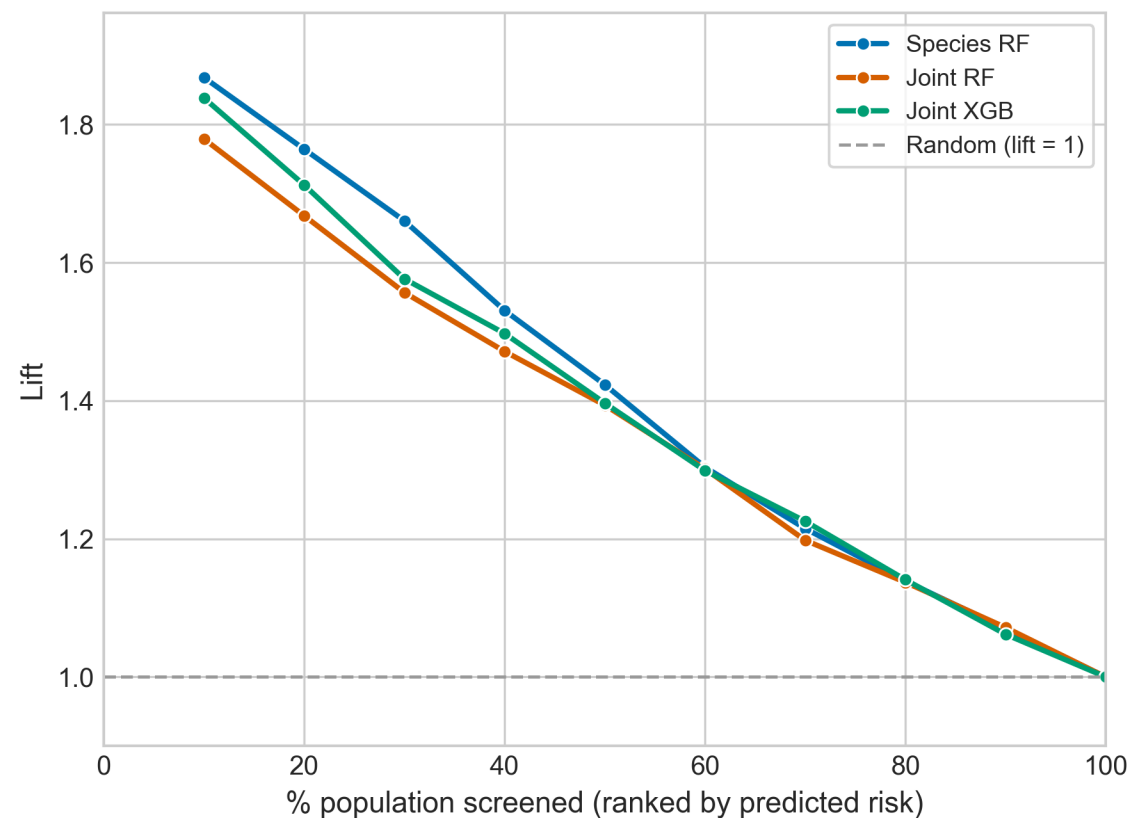


Cohort-exclusion sensitivity: 10-cohort headline vs 11-cohort (HanniganGD_2017 included)

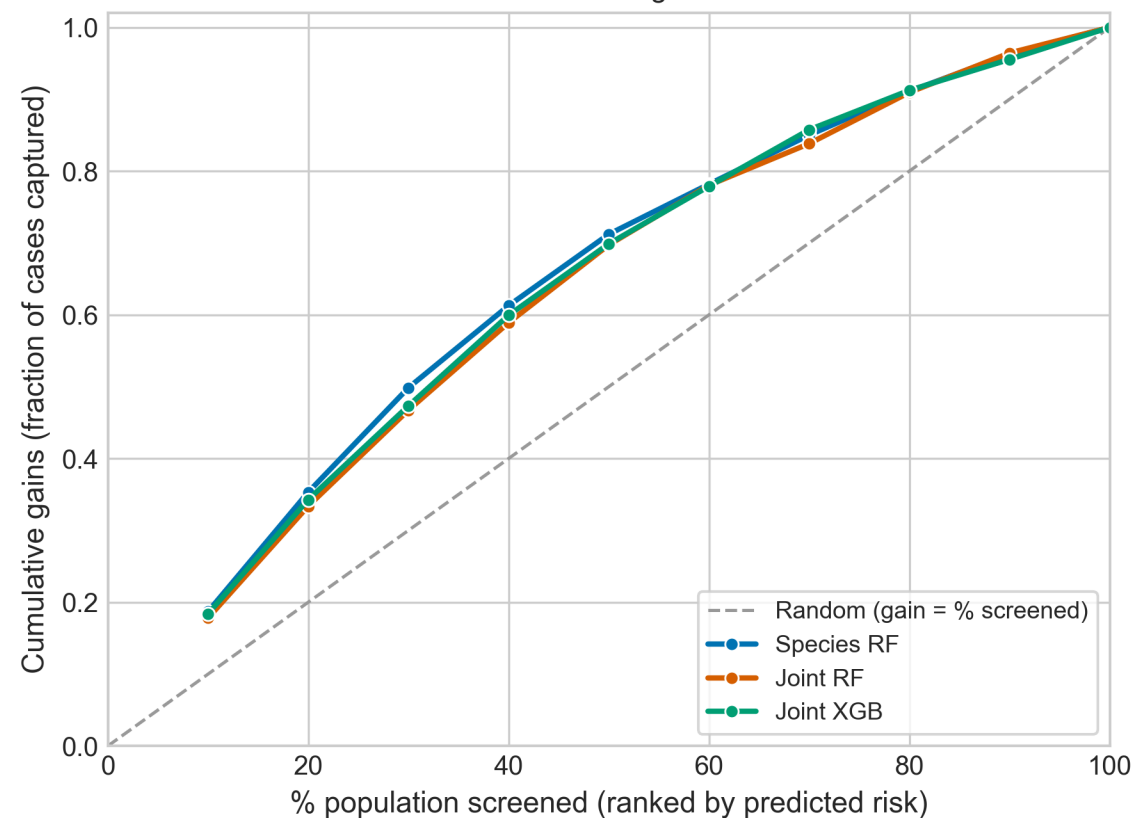


Lift and cumulative gains (pooled LODO predictions, n = 1339)

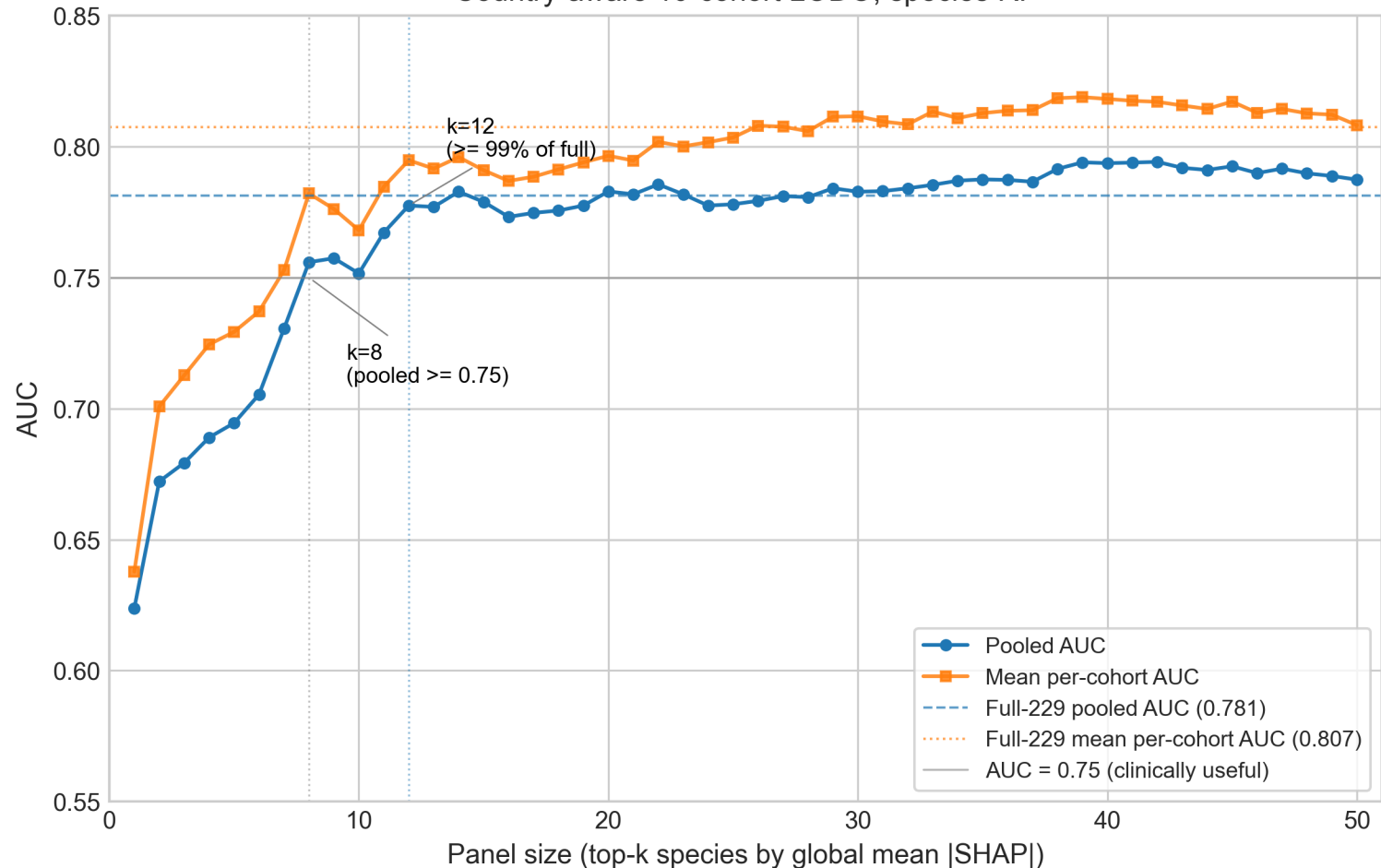
Lift curve



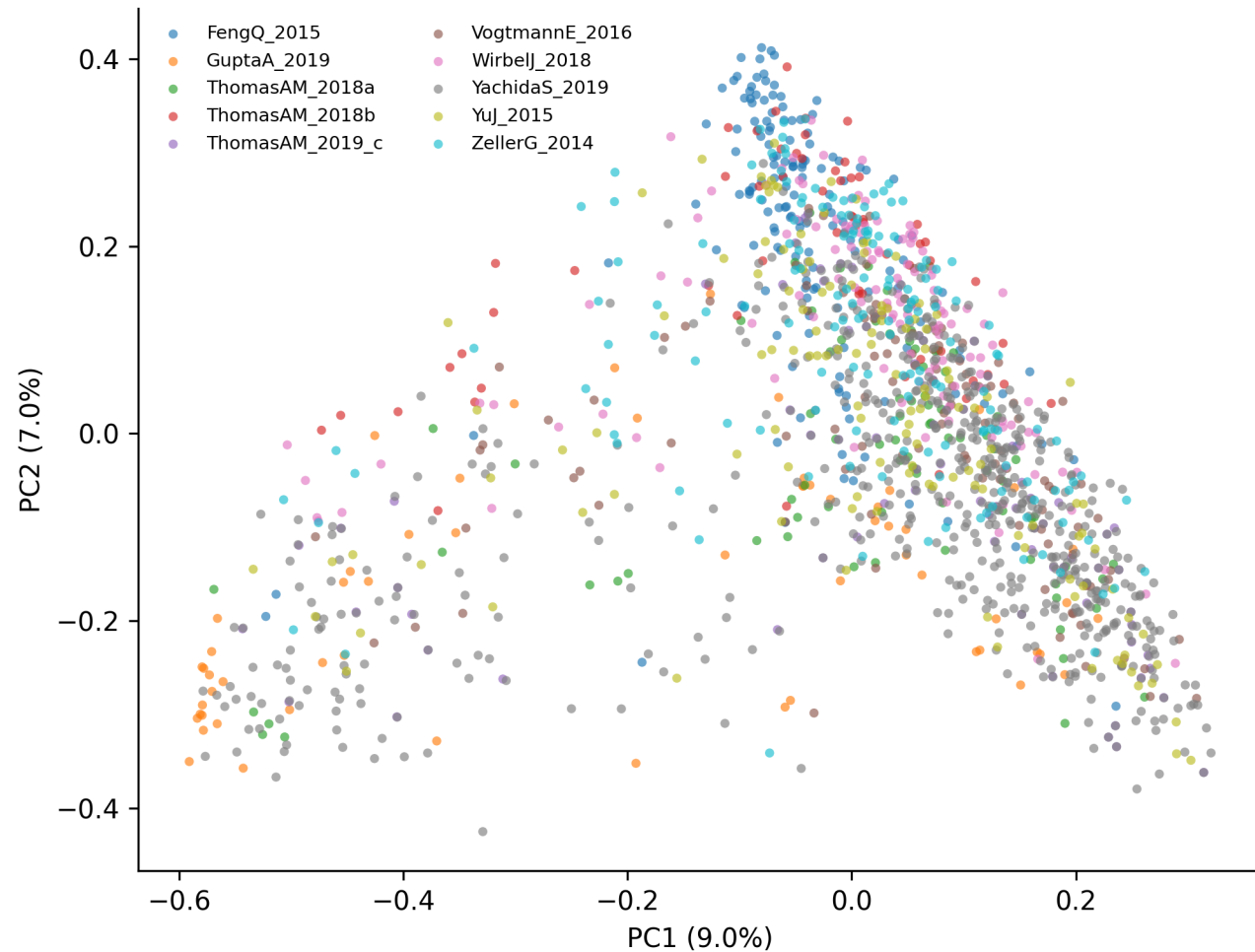
Cumulative gains



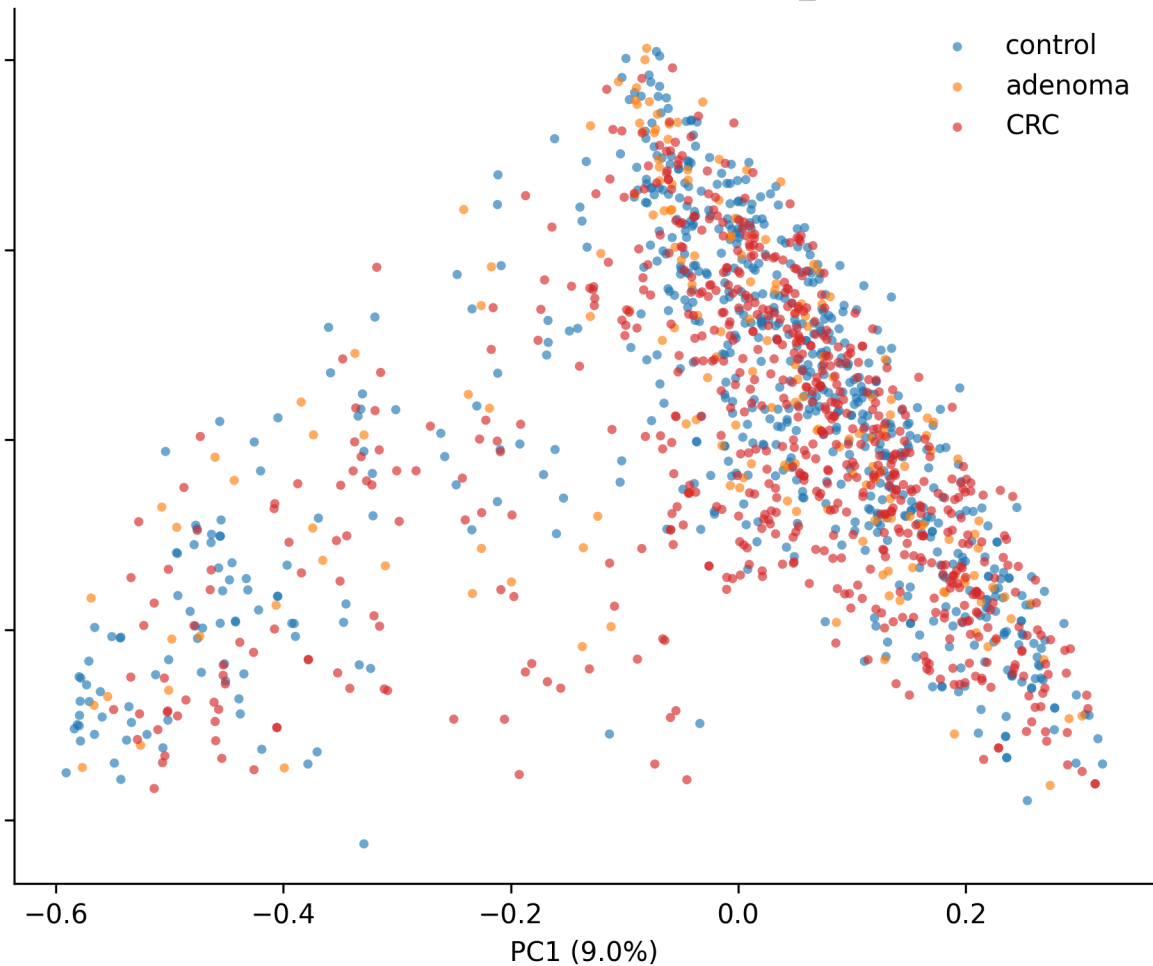
Minimum useful species panel — greedy forward selection
Country-aware 10-cohort LODO, species RF



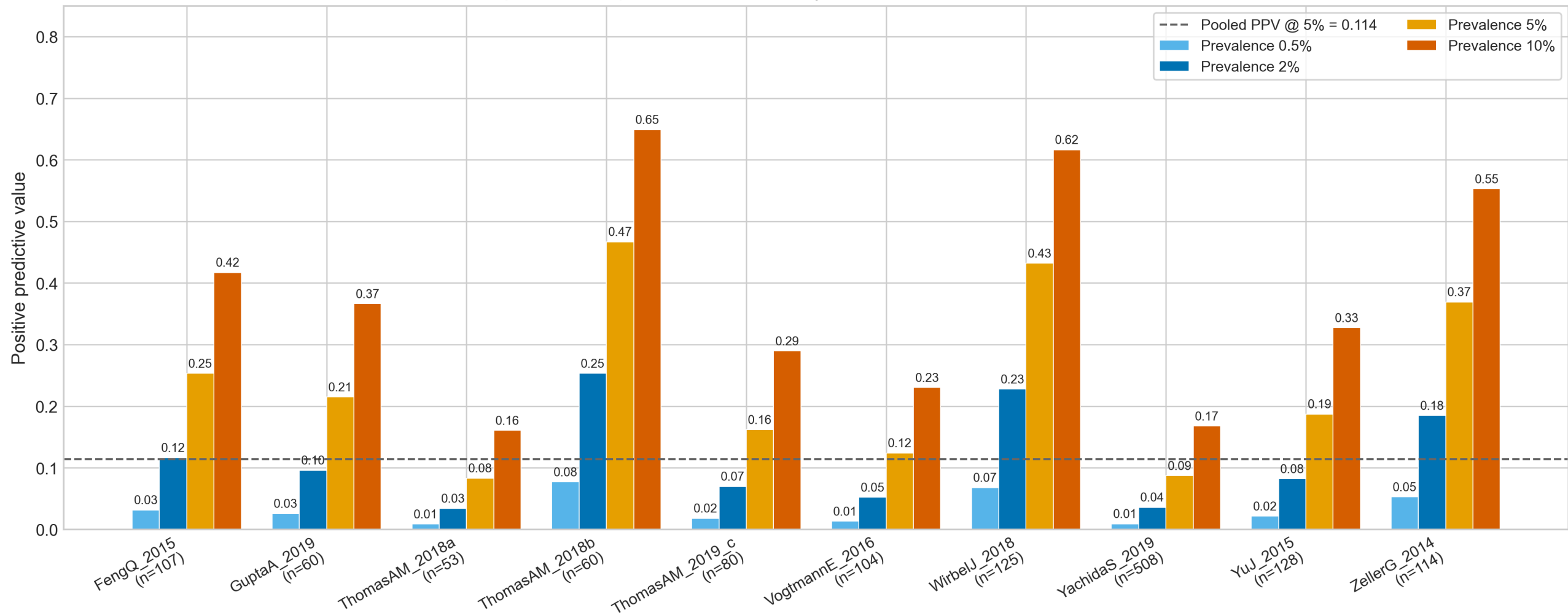
PCoA (Bray-Curtis) coloured by cohort

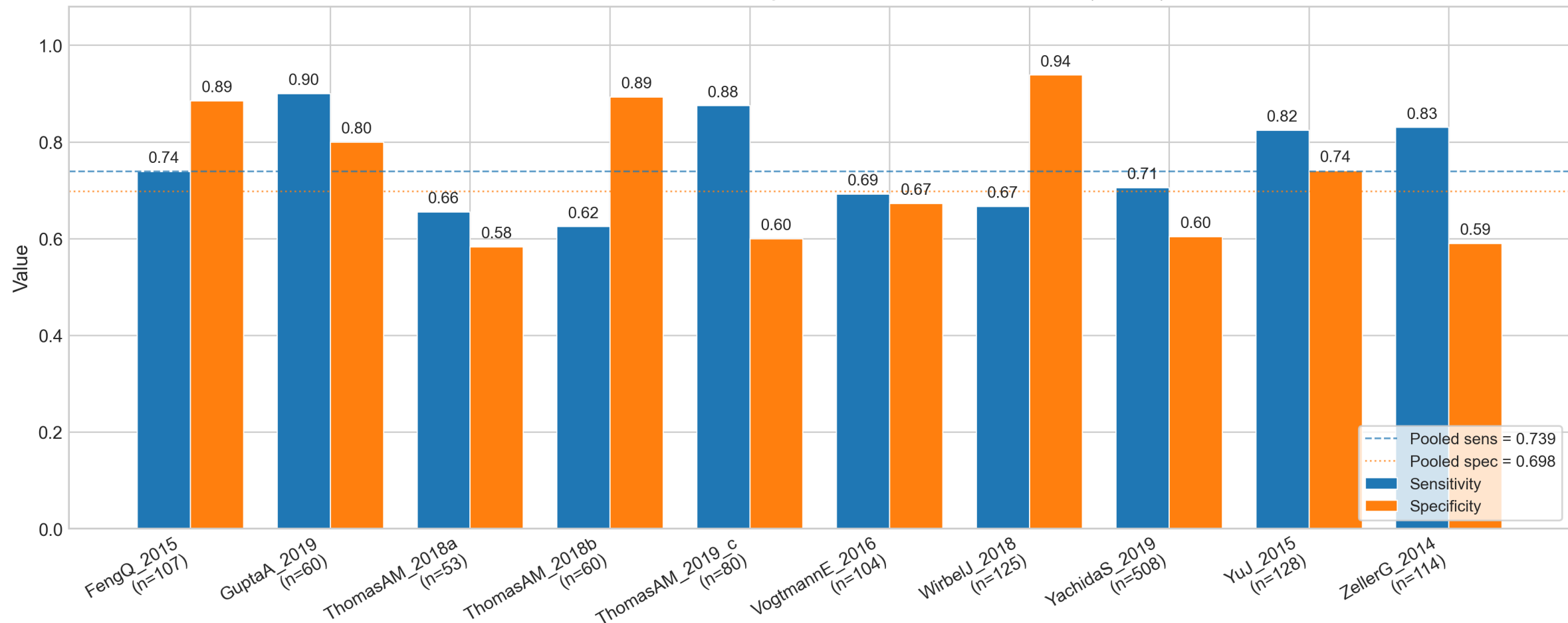


PCoA (Bray-Curtis) coloured by study_condition



Species RF — per-cohort base-rate-adjusted PPV at local Youden-J threshold

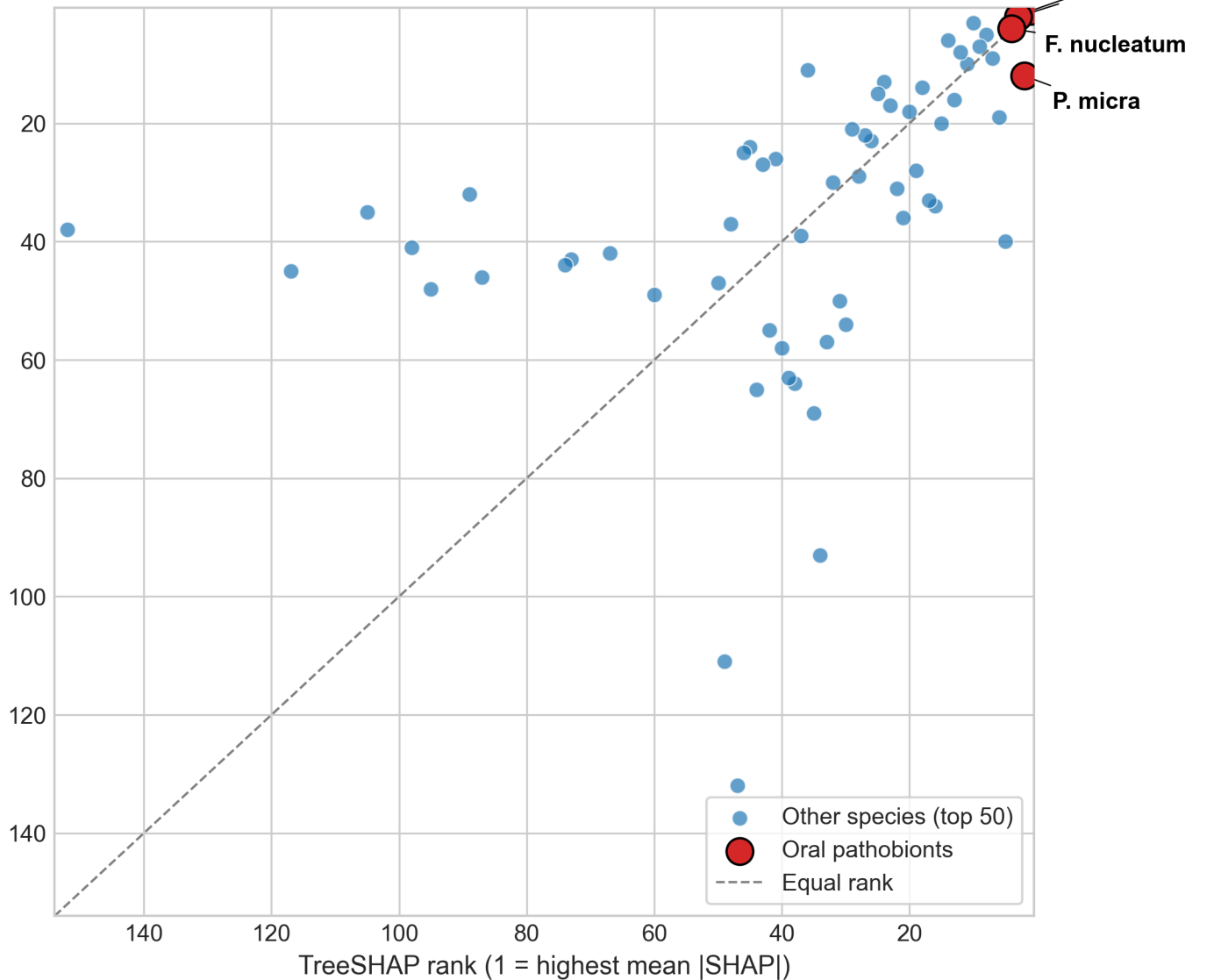


Species RF — per-cohort operating characteristics at pooled Youden-J ($t=0.485$)

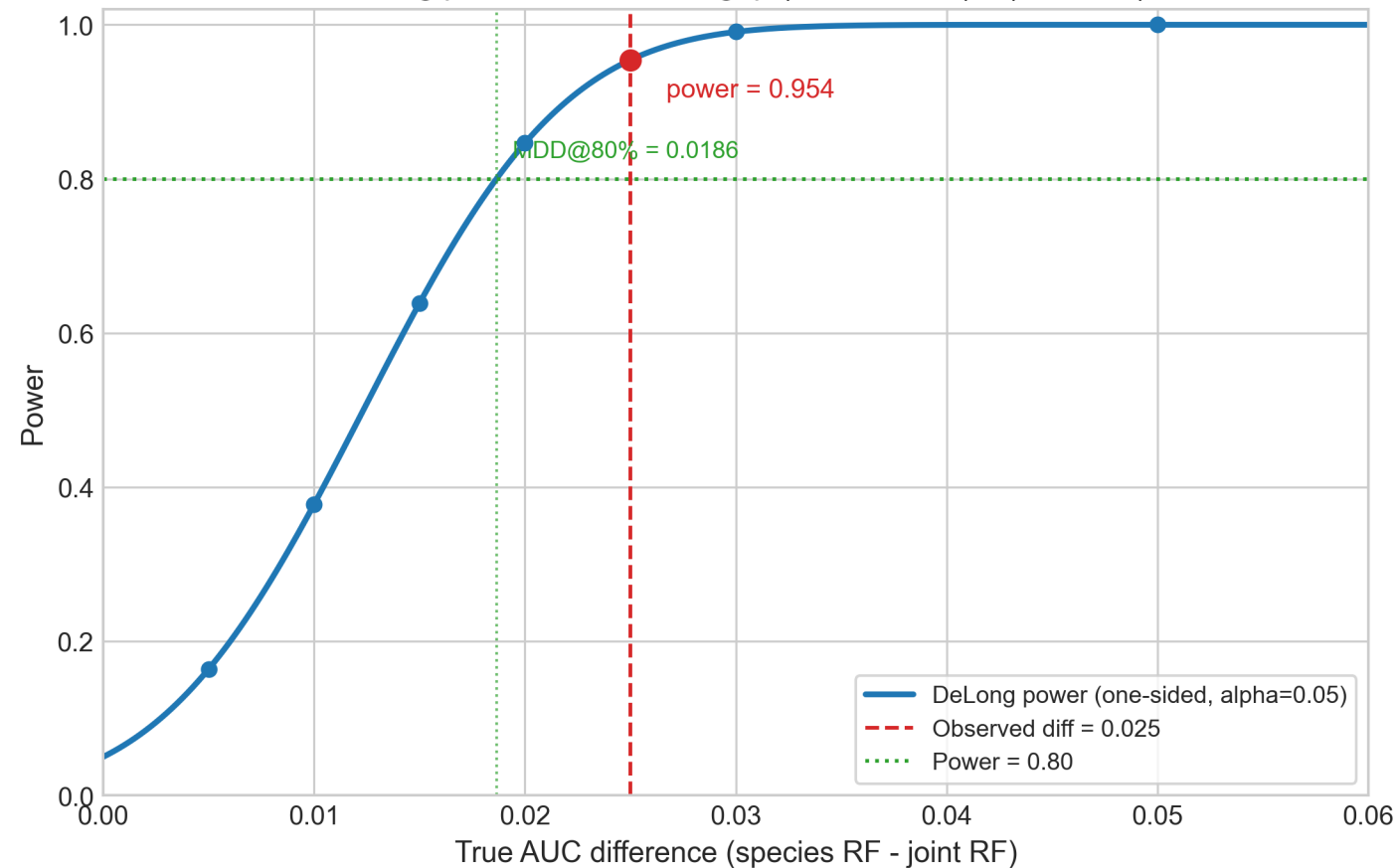
SHAP vs permutation importance — species RF, CRC vs control

Top-20 overlap: 16/20

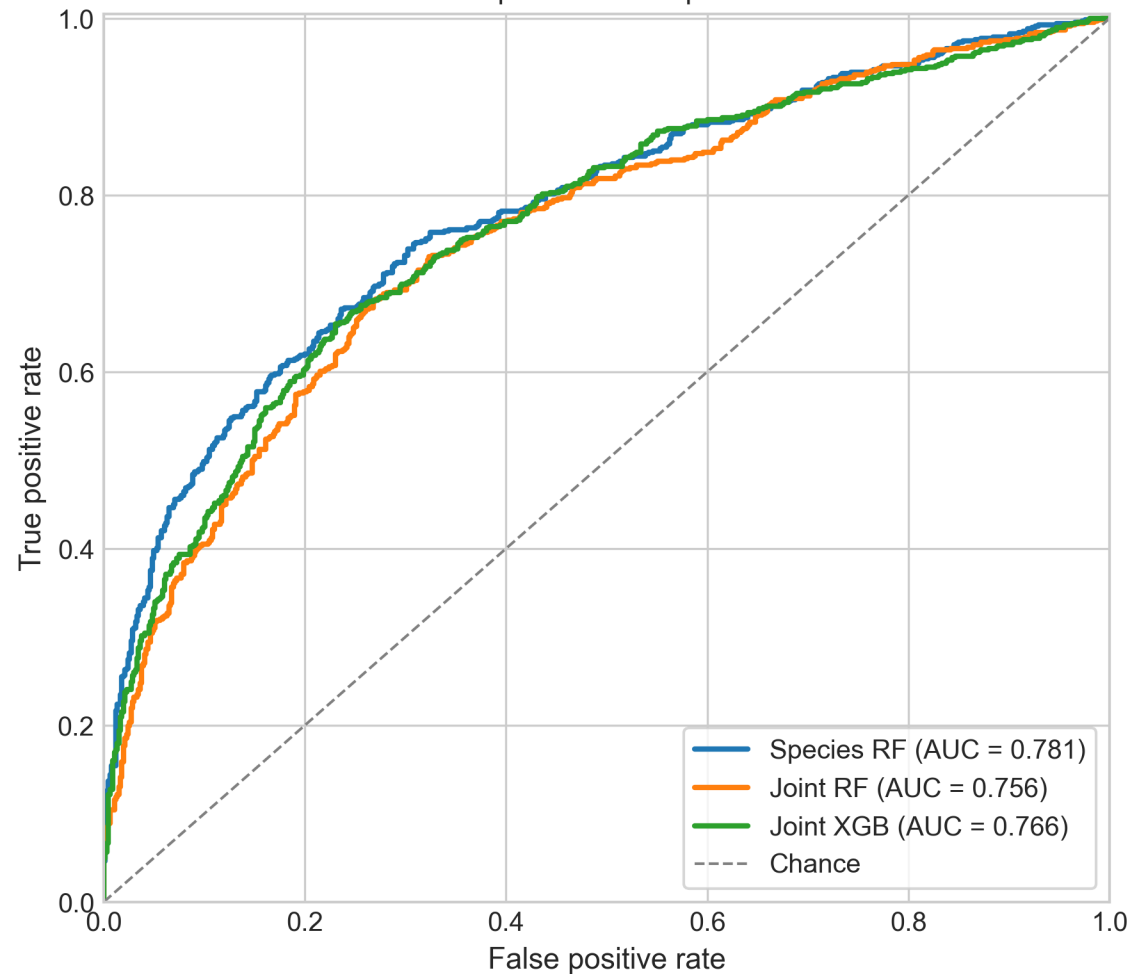
Permutation-importance rank (1 = largest AUC drop)



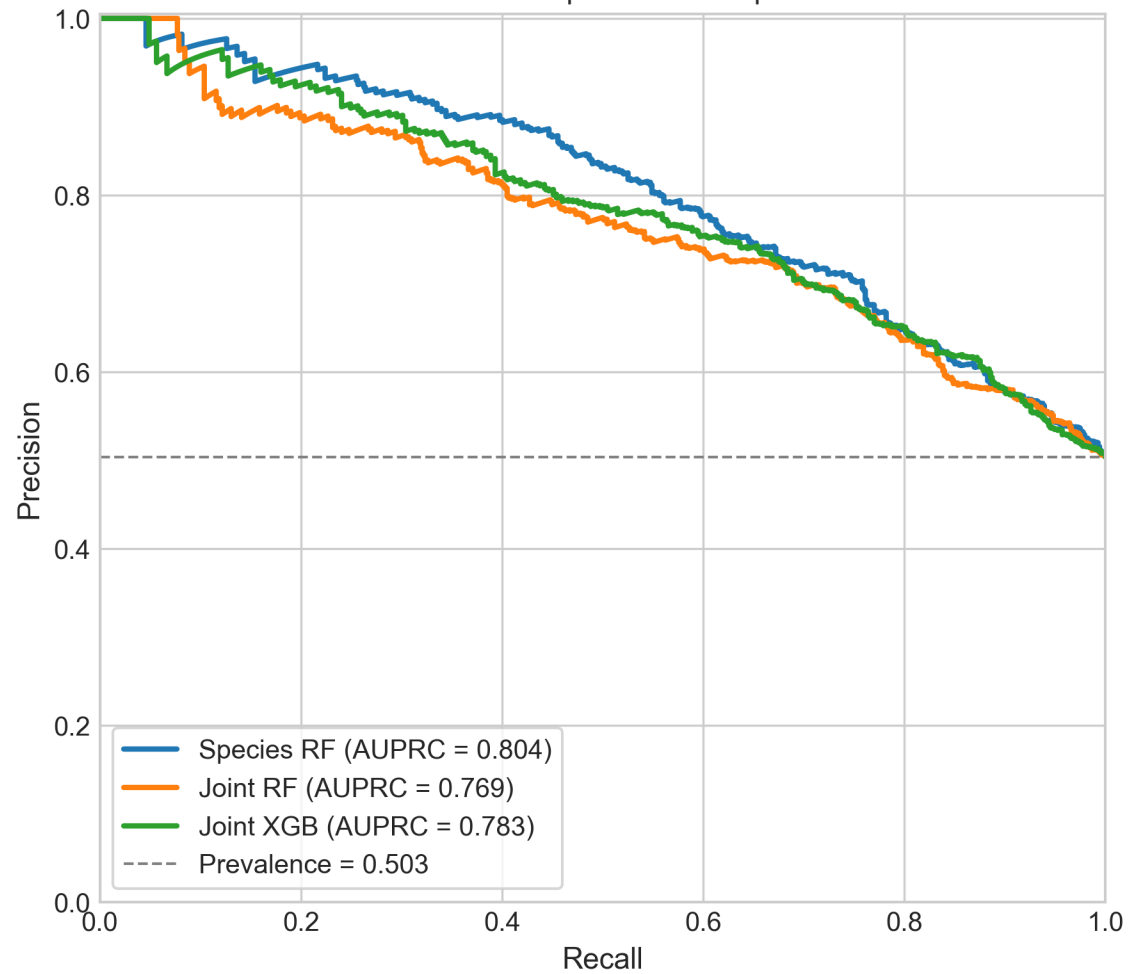
DeLong power vs true AUC gap (n=1339, SE(diff)=0.0075)



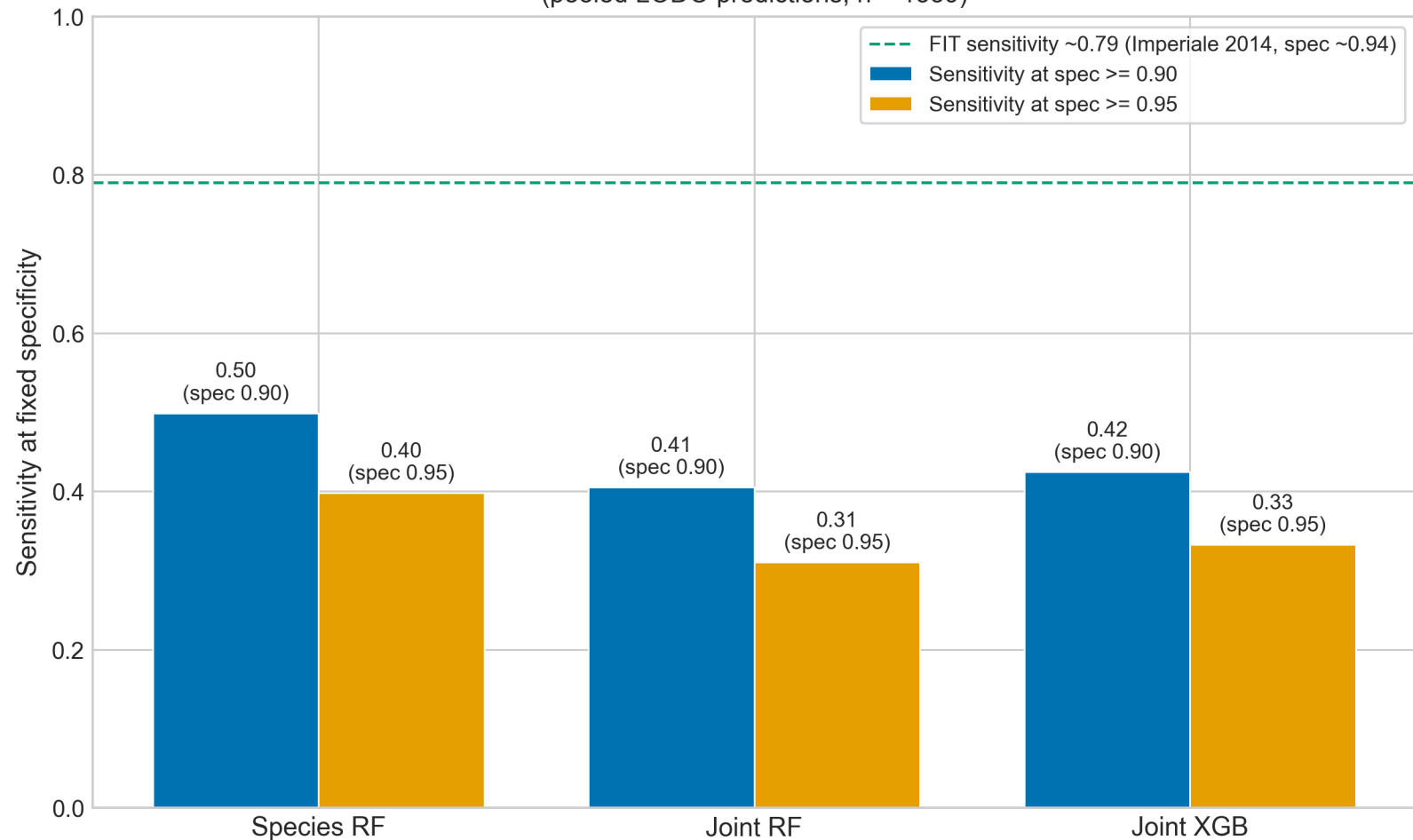
ROC — pooled LODO predictions



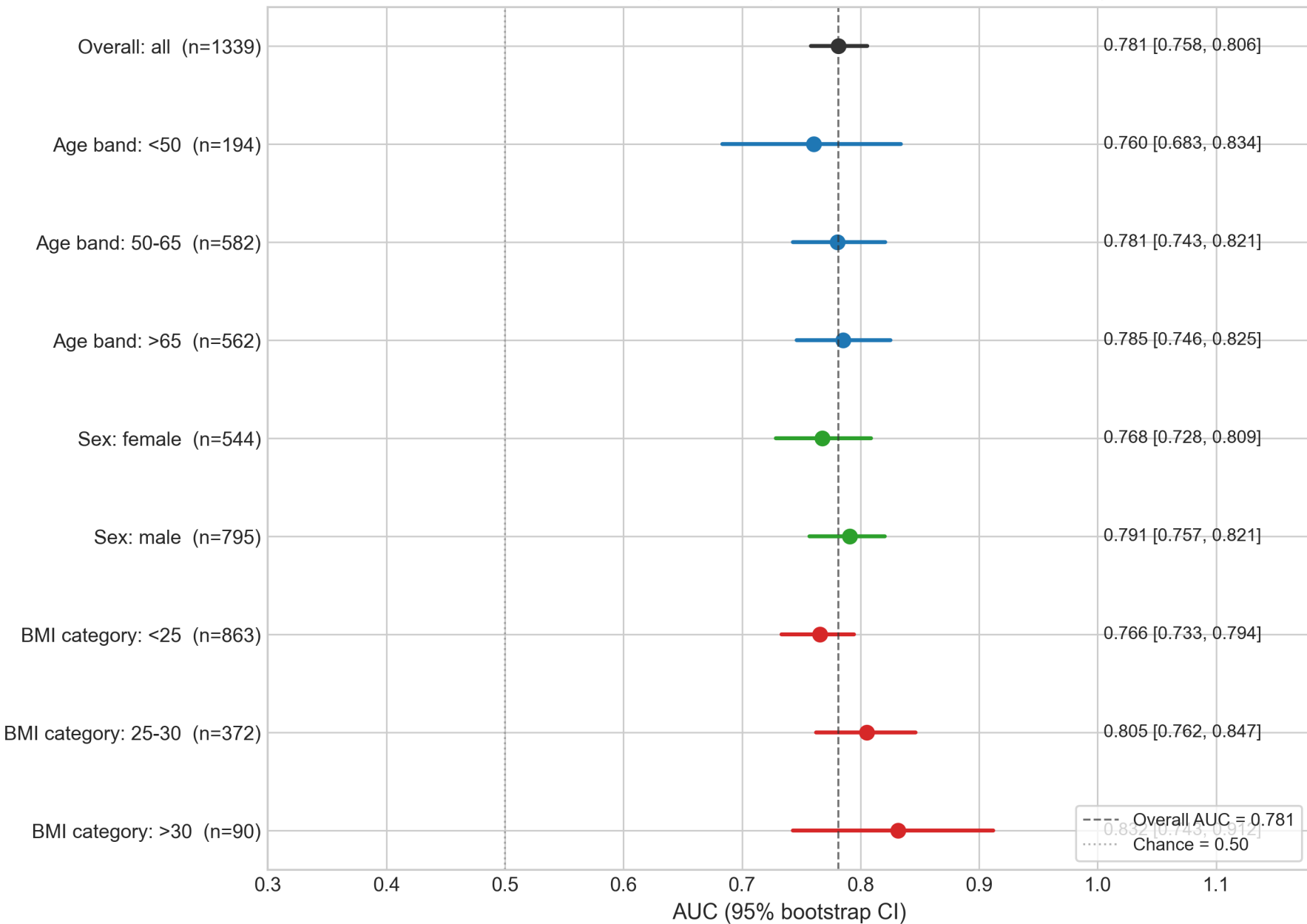
Precision-Recall — pooled LODO predictions



Sensitivity at clinically-meaningful specificity floors
(pooled LODO predictions, n = 1339)



Species RF — subgroup AUC (pooled LODO, 1000-iter bootstrap)



Top-20 most-abundant species (overall) by cohort

